

benchmarks

June 2, 2026

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[2]: !mkdir -p benchmark-results
!cd .. && timeout 120s ./build-release/bin/benchmarks
↳--benchmark_filter='^BM_SQL<(InterpretedExpressionExecutor|CachedJitCompiledExpressionExecu
↳(kSimpleSelectSmall|kJoinSmall|kComplex5), PlannerMode::(kNaive|kOptimized)>/
↳real_time$' --benchmark_repetitions=3
↳--benchmark_display_aggregates_only=true --benchmark_out=research/
↳benchmark-results/query.json --benchmark_out_format=json
```

2026-06-02T01:16:13+03:00

Running ./build-release/bin/benchmarks

Run on (8 X 4200 MHz CPU s)

CPU Caches:

L1 Data 48 KiB (x4)

L1 Instruction 32 KiB (x4)

L2 Unified 1280 KiB (x4)

L3 Unified 8192 KiB (x1)

Load Average: 0.38, 0.38, 0.46

WARNING CPU scaling is enabled, the benchmark real time measurements may be noisy and will incur extra overhead.

Benchmark

Time	CPU	Iterations
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BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall, PlannerMode::kNaive>/real_time_mean	22492 ns	
22349 ns	3	
BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall, PlannerMode::kNaive>/real_time_median	22447 ns	
22296 ns	3	
BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall, PlannerMode::kNaive>/real_time_stddev	1558 ns	
1526 ns	3	

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BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kNaive>/real_time_cv          6.93 %
6.83 %          3
BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_mean      20696 ns
20601 ns          3
BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_median     20835 ns
20742 ns          3
BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_stddev      834 ns
810 ns          3
BM_SQL<InterpretedExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_cv          4.03 %
3.93 %          3
BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kNaive>/real_time_mean           20039 ns          19954
ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kNaive>/real_time_median          19980 ns          19905
ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kNaive>/real_time_stddev           172 ns          177
ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kNaive>/real_time_cv              0.86 %          0.89 %
3
BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_mean          19665 ns          19591
ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_median          19751 ns          19677
ns          3

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BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_stddev          207 ns          204
ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kSimpleSelectSmall,
PlannerMode::kOptimized>/real_time_cv              1.05 %          1.04 %
3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_mean                71494 ns
70618 ns          3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_median              72186 ns
71298 ns          3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_stddev              3465 ns
3354 ns          3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_cv                  4.85 %
4.75 %          3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_mean            71193 ns
70379 ns          3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_median           71837 ns
70986 ns          3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_stddev           1447 ns
1398 ns          3
BM_SQL<InterpretedExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_cv               2.03 %
1.99 %          3
BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_mean                74523 ns
73518 ns          3

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BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_median          74232 ns
73228 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_stddev          3582 ns
3406 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kNaive>/real_time_cv              4.81 %
4.63 %          3
BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_mean        70604 ns
69783 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_median      68387 ns
67663 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_stddev      6690 ns
6389 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kJoinSmall,
PlannerMode::kOptimized>/real_time_cv          9.48 %
9.16 %          3
BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_mean            88746 ns
87780 ns          3
BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_median          87787 ns
86834 ns          3
BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_stddev          4903 ns
4717 ns          3
BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_cv              5.52 %
5.37 %          3

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BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_mean          93122 ns
92033 ns          3
BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_median          94811 ns
93696 ns          3
BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_stddev          3892 ns
3747 ns          3
BM_SQL<InterpretedExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_cv          4.18 %
4.07 %          3
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_mean          84456 ns
83477 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_median          85501 ns
84546 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_stddev          3598 ns
3410 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kNaive>/real_time_cv          4.26 %
4.08 %          3
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_mean          80405 ns
79621 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_median          80566 ns
79795 ns          3
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_stddev          536 ns
528 ns          3

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```
BM_SQL<CachedJitCompiledExpressionExecutor, kComplex5,
PlannerMode::kOptimized>/real_time_cv          0.67 %
0.66 %          3
```

```
[3]: !cd .. && SSB_DATA_DIR=benchmarks/datasets/ssb/generated/sf001 timeout 180s ./
↳ build-release/bin/benchmarks --benchmark_filter='^SSB/'
↳ --benchmark_repetitions=3 --benchmark_display_aggregates_only=true
↳ --benchmark_out=research/benchmark-results/ssb-sf001.json
↳ --benchmark_out_format=json
```

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Running ./build-release/bin/benchmarks

Run on (8 X 4200 MHz CPU s)

CPU Caches:

L1 Data 48 KiB (x4)

L1 Instruction 32 KiB (x4)

L2 Unified 1280 KiB (x4)

L3 Unified 8192 KiB (x1)

Load Average: 0.80, 0.49, 0.49

WARNING CPU scaling is enabled, the benchmark real time measurements may be noisy and will incur extra overhead.

Benchmark	Time	CPU
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SSB/q2.1/Interpreted/Naive/real_time_mean	9535917 ns	
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9457917 ns 3 execution_us=9.53583k

includes_order_by=1

SSB/q2.1/Interpreted/Naive/real_time_median	9539993	
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ns 9459233 ns 3 execution_us=9.53993k

includes_order_by=1

SSB/q2.1/Interpreted/Naive/real_time_stddev	26903	
---	-------	--

ns 26949 ns 3 execution_us=26.9284

includes_order_by=0

SSB/q2.1/Interpreted/Naive/real_time_cv	0.28 %	
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0.28 % 3 execution_us=0.28%

includes_order_by=0.00%

SSB/q2.1/Interpreted/Optimized/real_time_mean	10018523	
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ns 9946935 ns 3 execution_us=10.0184k

includes_order_by=1 optimizer_us=818.029 plan_cost=836.603k

SSB/q2.1/Interpreted/Optimized/real_time_median 10025189

ns 9956589 ns 3 execution_us=10.0251k
includes_order_by=1 optimizer_us=797.011 plan_cost=836.603k

SSB/q2.1/Interpreted/Optimized/real_time_stddev 23507

ns 26853 ns 3 execution_us=23.4951
includes_order_by=0 optimizer_us=47.3596 plan_cost=0

SSB/q2.1/Interpreted/Optimized/real_time_cv 0.23 %

0.27 % 3 execution_us=0.23%
includes_order_by=0.00% optimizer_us=5.79% plan_cost=0.00%

SSB/q1.3/Interpreted/Naive/real_time_mean 8894810

ns 8830456 ns 3 execution_us=8.89474k
includes_order_by=0

SSB/q1.3/Interpreted/Naive/real_time_median 8703923

ns 8643932 ns 3 execution_us=8.70386k
includes_order_by=0

SSB/q1.3/Interpreted/Naive/real_time_stddev 370836

ns 362794 ns 3 execution_us=370.843
includes_order_by=0

SSB/q1.3/Interpreted/Naive/real_time_cv 4.17 %

4.11 % 3 execution_us=4.17%
includes_order_by=0.00%

SSB/q1.3/Interpreted/Optimized/real_time_mean 7216482

ns 7178427 ns 3 execution_us=7.21641k
includes_order_by=0 optimizer_us=419.653 plan_cost=654.297k

SSB/q1.3/Interpreted/Optimized/real_time_median 7171734

ns 7135746 ns 3 execution_us=7.17166k
includes_order_by=0 optimizer_us=404.975 plan_cost=654.297k

SSB/q1.3/Interpreted/Optimized/real_time_stddev 378775

ns 369597 ns 3 execution_us=378.779
includes_order_by=0 optimizer_us=40.6977 plan_cost=0

SSB/q1.3/Interpreted/Optimized/real_time_cv 5.25 %

5.15 % 3 execution_us=5.25%
includes_order_by=0.00% optimizer_us=9.70% plan_cost=0.00%

SSB/q1.2/Interpreted/Naive/real_time_mean 8755501

ns 8695939 ns 3 execution_us=8.75534k
includes_order_by=0

SSB/q1.2/Interpreted/Naive/real_time_median 8634674

ns 8580077 ns 3 execution_us=8.63449k
includes_order_by=0

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SSB/q1.2/Interpreted/Naive/real_time_stddev      313421
ns          306308 ns          3 execution_us=313.48
includes_order_by=0
SSB/q1.2/Interpreted/Naive/real_time_cv          3.58 %
3.52 %          3 execution_us=3.58%
includes_order_by=0.00%
SSB/q1.2/Interpreted/Optimized/real_time_mean     6842163
ns          6813610 ns          3 execution_us=6.8421k
includes_order_by=0 optimizer_us=225.618 plan_cost=654.297k
SSB/q1.2/Interpreted/Optimized/real_time_median   6857864
ns          6828042 ns          3 execution_us=6.8578k
includes_order_by=0 optimizer_us=225.797 plan_cost=654.297k
SSB/q1.2/Interpreted/Optimized/real_time_stddev    47976
ns          47617 ns          3 execution_us=47.9819
includes_order_by=0 optimizer_us=2.58966 plan_cost=0
SSB/q1.2/Interpreted/Optimized/real_time_cv        0.70 %
0.70 %          3 execution_us=0.70%
includes_order_by=0.00% optimizer_us=1.15% plan_cost=0.00%
SSB/q3.3/Interpreted/Naive/real_time_mean         38770551
ns          38545894 ns          3 execution_us=38.7704k
includes_order_by=1
SSB/q3.3/Interpreted/Naive/real_time_median       38937311
ns          38712070 ns          3 execution_us=38.9372k
includes_order_by=1
SSB/q3.3/Interpreted/Naive/real_time_stddev       403708
ns          387320 ns          3 execution_us=403.68
includes_order_by=0
SSB/q3.3/Interpreted/Naive/real_time_cv           1.04 %
1.00 %          3 execution_us=1.04%
includes_order_by=0.00%
SSB/q3.3/Interpreted/Optimized/real_time_mean     6165384
ns          6131574 ns          3 execution_us=6.16533k
includes_order_by=1 optimizer_us=1.97412k plan_cost=817.083k
SSB/q3.3/Interpreted/Optimized/real_time_median   6220261
ns          6184963 ns          3 execution_us=6.2202k
includes_order_by=1 optimizer_us=1.92956k plan_cost=817.083k
SSB/q3.3/Interpreted/Optimized/real_time_stddev    109818
ns          106871 ns          3 execution_us=109.811
includes_order_by=0 optimizer_us=127.329 plan_cost=0

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SSB/q3.3/Interpreted/Optimized/real_time_cv 1.78 %
1.74 % 3 execution_us=1.78%
includes_order_by=0.00% optimizer_us=6.45% plan_cost=0.00%
SSB/q3.4/Interpreted/Naive/real_time_mean 38014861
ns 37785646 ns 3 execution_us=38.0147k
includes_order_by=1
SSB/q3.4/Interpreted/Naive/real_time_median 37373498
ns 37160567 ns 3 execution_us=37.3733k
includes_order_by=1
SSB/q3.4/Interpreted/Naive/real_time_stddev 1152550
ns 1124559 ns 3 execution_us=1.15257k
includes_order_by=0
SSB/q3.4/Interpreted/Naive/real_time_cv 3.03 %
2.98 % 3 execution_us=3.03%
includes_order_by=0.00%
SSB/q3.4/Interpreted/Optimized/real_time_mean 6548677
ns 6491466 ns 3 execution_us=6.54861k
includes_order_by=1 optimizer_us=1.73735k plan_cost=817.083k
SSB/q3.4/Interpreted/Optimized/real_time_median 6752747
ns 6702527 ns 3 execution_us=6.75267k
includes_order_by=1 optimizer_us=1.76762k plan_cost=817.083k
SSB/q3.4/Interpreted/Optimized/real_time_stddev 547461
ns 516050 ns 3 execution_us=547.457
includes_order_by=0 optimizer_us=55.8846 plan_cost=0
SSB/q3.4/Interpreted/Optimized/real_time_cv 8.36 %
7.95 % 3 execution_us=8.36%
includes_order_by=0.00% optimizer_us=3.22% plan_cost=0.00%
SSB/q4.3/Interpreted/Naive/real_time_mean 9839082
ns 9773621 ns 3 execution_us=9.83901k
includes_order_by=1
SSB/q4.3/Interpreted/Naive/real_time_median 9830951
ns 9757826 ns 3 execution_us=9.83088k
includes_order_by=1
SSB/q4.3/Interpreted/Naive/real_time_stddev 29124
ns 30559 ns 3 execution_us=29.1302
includes_order_by=0
SSB/q4.3/Interpreted/Naive/real_time_cv 0.30 %
0.31 % 3 execution_us=0.30%
includes_order_by=0.00%

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SSB/q4.3/Interpreted/Optimized/real_time_mean      6726511
ns          6685741 ns          3 execution_us=6.72644k
includes_order_by=1 optimizer_us=3.74304k plan_cost=841.733k
SSB/q4.3/Interpreted/Optimized/real_time_median     6733679
ns          6696410 ns          3 execution_us=6.73358k
includes_order_by=1 optimizer_us=3.75501k plan_cost=841.733k
SSB/q4.3/Interpreted/Optimized/real_time_stddev      20124
ns          21507 ns          3 execution_us=20.1209
includes_order_by=0 optimizer_us=100.716 plan_cost=0
SSB/q4.3/Interpreted/Optimized/real_time_cv          0.30 %
0.32 %          3 execution_us=0.30%
includes_order_by=0.00% optimizer_us=2.69% plan_cost=0.00%
SSB/q1.1/Interpreted/Naive/real_time_mean           8337287
ns          8286855 ns          3 execution_us=8.33722k
includes_order_by=0
SSB/q1.1/Interpreted/Naive/real_time_median          8332199
ns          8279335 ns          3 execution_us=8.33213k
includes_order_by=0
SSB/q1.1/Interpreted/Naive/real_time_stddev          30591
ns          33255 ns          3 execution_us=30.5828
includes_order_by=0
SSB/q1.1/Interpreted/Naive/real_time_cv             0.37 %
0.40 %          3 execution_us=0.37%
includes_order_by=0.00%
SSB/q1.1/Interpreted/Optimized/real_time_mean       7009090
ns          6972349 ns          3 execution_us=7.00901k
includes_order_by=0 optimizer_us=203.094 plan_cost=705.057k
SSB/q1.1/Interpreted/Optimized/real_time_median     7011279
ns          6975684 ns          3 execution_us=7.01123k
includes_order_by=0 optimizer_us=203.772 plan_cost=705.057k
SSB/q1.1/Interpreted/Optimized/real_time_stddev      7876
ns          7272 ns          3 execution_us=7.88332
includes_order_by=0 optimizer_us=12.9813 plan_cost=0
SSB/q1.1/Interpreted/Optimized/real_time_cv          0.11 %
0.10 %          3 execution_us=0.11%
includes_order_by=0.00% optimizer_us=6.39% plan_cost=0.00%
SSB/q3.2/Interpreted/Naive/real_time_mean           41168218
ns          40861121 ns          3 execution_us=41.168k
includes_order_by=1

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SSB/q3.2/Interpreted/Naive/real_time_median      39951647
ns      39624417 ns      3 execution_us=39.9514k
includes_order_by=1
SSB/q3.2/Interpreted/Naive/real_time_stddev      2772999
ns      2719430 ns      3 execution_us=2.77305k
includes_order_by=0
SSB/q3.2/Interpreted/Naive/real_time_cv          6.74 %
6.66 %      3 execution_us=6.74%
includes_order_by=0.00%
SSB/q3.2/Interpreted/Optimized/real_time_mean      7331045
ns      7263440 ns      3 execution_us=7.33098k
includes_order_by=1 optimizer_us=1.77516k plan_cost=816.743k
SSB/q3.2/Interpreted/Optimized/real_time_median      7250476
ns      7179529 ns      3 execution_us=7.25042k
includes_order_by=1 optimizer_us=1.45913k plan_cost=816.743k
SSB/q3.2/Interpreted/Optimized/real_time_stddev      560067
ns      542530 ns      3 execution_us=560.073
includes_order_by=0 optimizer_us=631.618 plan_cost=0
SSB/q3.2/Interpreted/Optimized/real_time_cv          7.64 %
7.47 %      3 execution_us=7.64%
includes_order_by=0.00% optimizer_us=35.58% plan_cost=0.00%
SSB/q2.2/Interpreted/Naive/real_time_mean      10611863
ns      10523713 ns      3 execution_us=10.6118k
includes_order_by=1
SSB/q2.2/Interpreted/Naive/real_time_median      10571693
ns      10485617 ns      3 execution_us=10.5716k
includes_order_by=1
SSB/q2.2/Interpreted/Naive/real_time_stddev      700693
ns      680849 ns      3 execution_us=700.727
includes_order_by=0
SSB/q2.2/Interpreted/Naive/real_time_cv          6.60 %
6.47 %      3 execution_us=6.60%
includes_order_by=0.00%
SSB/q2.2/Interpreted/Optimized/real_time_mean      7990721
ns      7922960 ns      3 execution_us=7.99066k
includes_order_by=1 optimizer_us=1.30379k plan_cost=841.703k
SSB/q2.2/Interpreted/Optimized/real_time_median      7968509
ns      7901081 ns      3 execution_us=7.96844k
includes_order_by=1 optimizer_us=1.07317k plan_cost=841.703k

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SSB/q2.2/Interpreted/Optimized/real_time_stddev      412217
ns          399538 ns          3 execution_us=412.219
includes_order_by=0 optimizer_us=435.934 plan_cost=0
SSB/q2.2/Interpreted/Optimized/real_time_cv          5.16 %
5.04 %          3 execution_us=5.16%
includes_order_by=0.00% optimizer_us=33.44% plan_cost=0.00%
SSB/q3.1/Interpreted/Naive/real_time_mean          41047048
ns          40776811 ns          3 execution_us=41.0468k
includes_order_by=1
SSB/q3.1/Interpreted/Naive/real_time_median          40607061
ns          40351481 ns          3 execution_us=40.6066k
includes_order_by=1
SSB/q3.1/Interpreted/Naive/real_time_stddev          1022743
ns          1007568 ns          3 execution_us=1.02279k
includes_order_by=0
SSB/q3.1/Interpreted/Naive/real_time_cv          2.49 %
2.47 %          3 execution_us=2.49%
includes_order_by=0.00%
SSB/q3.1/Interpreted/Optimized/real_time_mean        7329905
ns          7261047 ns          3 execution_us=7.32985k
includes_order_by=1 optimizer_us=1.56312k plan_cost=816.743k
SSB/q3.1/Interpreted/Optimized/real_time_median        7444323
ns          7376360 ns          3 execution_us=7.44426k
includes_order_by=1 optimizer_us=1.60169k plan_cost=816.743k
SSB/q3.1/Interpreted/Optimized/real_time_stddev        377349
ns          367470 ns          3 execution_us=377.353
includes_order_by=0 optimizer_us=243.913 plan_cost=0
SSB/q3.1/Interpreted/Optimized/real_time_cv          5.15 %
5.06 %          3 execution_us=5.15%
includes_order_by=0.00% optimizer_us=15.60% plan_cost=0.00%
SSB/q4.2/Interpreted/Naive/real_time_mean          11383627
ns          11273121 ns          3 execution_us=11.3835k
includes_order_by=1
SSB/q4.2/Interpreted/Naive/real_time_median          11182684
ns          11075830 ns          3 execution_us=11.1826k
includes_order_by=1
SSB/q4.2/Interpreted/Naive/real_time_stddev          462093
ns          441146 ns          3 execution_us=462.11
includes_order_by=0

```

```

SSB/q4.2/Interpreted/Naive/real_time_cv          4.06 %
3.91 %          3 execution_us=4.06%
includes_order_by=0.00%
SSB/q4.2/Interpreted/Optimized/real_time_mean    13753622
ns          13641036 ns          3 execution_us=13.7534k
includes_order_by=1 optimizer_us=6.6919k plan_cost=843.203k
SSB/q4.2/Interpreted/Optimized/real_time_median  13773635
ns          13660823 ns          3 execution_us=13.7735k
includes_order_by=1 optimizer_us=6.40666k plan_cost=843.203k
SSB/q4.2/Interpreted/Optimized/real_time_stddev   1461768
ns          1423624 ns          3 execution_us=1.46187k
includes_order_by=0 optimizer_us=859.324 plan_cost=0
SSB/q4.2/Interpreted/Optimized/real_time_cv      10.63 %
10.44 %          3 execution_us=10.63%
includes_order_by=0.00% optimizer_us=12.84% plan_cost=0.00%
SSB/q4.1/Interpreted/Naive/real_time_mean        9885573
ns          9821993 ns          3 execution_us=9.88549k
includes_order_by=1
SSB/q4.1/Interpreted/Naive/real_time_median      9935466
ns          9868635 ns          3 execution_us=9.93537k
includes_order_by=1
SSB/q4.1/Interpreted/Naive/real_time_stddev      126113
ns          121567 ns          3 execution_us=126.119
includes_order_by=0
SSB/q4.1/Interpreted/Naive/real_time_cv          1.28 %
1.24 %          3 execution_us=1.28%
includes_order_by=0.00%
SSB/q4.1/Interpreted/Optimized/real_time_mean    13621071
ns          13528927 ns          3 execution_us=13.621k
includes_order_by=1 optimizer_us=3.96999k plan_cost=843.173k
SSB/q4.1/Interpreted/Optimized/real_time_median  13572908
ns          13484295 ns          3 execution_us=13.5728k
includes_order_by=1 optimizer_us=3.91961k plan_cost=843.173k
SSB/q4.1/Interpreted/Optimized/real_time_stddev   319283
ns          314861 ns          3 execution_us=319.283
includes_order_by=0 optimizer_us=115.745 plan_cost=0
SSB/q4.1/Interpreted/Optimized/real_time_cv      2.34 %
2.33 %          3 execution_us=2.34%
includes_order_by=0.00% optimizer_us=2.92% plan_cost=0.00%

```

```

SSB/q2.3/Interpreted/Naive/real_time_mean          9506406
ns          9448201 ns          3 execution_us=9.50634k
includes_order_by=1
SSB/q2.3/Interpreted/Naive/real_time_median          9494186
ns          9433948 ns          3 execution_us=9.49412k
includes_order_by=1
SSB/q2.3/Interpreted/Naive/real_time_stddev          72200
ns          71487 ns          3 execution_us=72.194
includes_order_by=0
SSB/q2.3/Interpreted/Naive/real_time_cv          0.76 %
0.76 %          3 execution_us=0.76%
includes_order_by=0.00%
SSB/q2.3/Interpreted/Optimized/real_time_mean          6827825
ns          6787123 ns          3 execution_us=6.82777k
includes_order_by=1 optimizer_us=852.138 plan_cost=836.603k
SSB/q2.3/Interpreted/Optimized/real_time_median          6823835
ns          6784459 ns          3 execution_us=6.82378k
includes_order_by=1 optimizer_us=847.591 plan_cost=836.603k
SSB/q2.3/Interpreted/Optimized/real_time_stddev          195671
ns          191222 ns          3 execution_us=195.664
includes_order_by=0 optimizer_us=23.0469 plan_cost=0
SSB/q2.3/Interpreted/Optimized/real_time_cv          2.87 %
2.82 %          3 execution_us=2.87%
includes_order_by=0.00% optimizer_us=2.70% plan_cost=0.00%

```

```

[4]: !cd .. && timeout 300s ./build-release/bin/benchmarks
      ↪--benchmark_filter='^OperatorCost/' --benchmark_repetitions=3
      ↪--benchmark_display_aggregates_only=true --benchmark_out=research/
      ↪benchmark-results/operator-cost.json --benchmark_out_format=json

```

2026-06-02T01:17:41+03:00

Running ./build-release/bin/benchmarks

Run on (8 X 4200 MHz CPU s)

CPU Caches:

L1 Data 48 KiB (x4)

L1 Instruction 32 KiB (x4)

L2 Unified 1280 KiB (x4)

L3 Unified 8192 KiB (x1)

Load Average: 1.43, 0.74, 0.58

WARNING CPU scaling is enabled, the benchmark real time measurements may be noisy and will incur extra overhead.

-----			Time
Benchmark	CPU	Iterations	UserCounters...

OperatorCost/SeqScan/1024/real_time_mean

38473 ns 38210 ns 3 model_cost=102.4k

output_rows=1.024k plan_cost=102.4k rows=1.024k

OperatorCost/SeqScan/1024/real_time_median

38428 ns 38240 ns 3 model_cost=102.4k

output_rows=1.024k plan_cost=102.4k rows=1.024k

OperatorCost/SeqScan/1024/real_time_stddev

753 ns 851 ns 3 model_cost=0

output_rows=0 plan_cost=0 rows=0

OperatorCost/SeqScan/1024/real_time_cv

1.96 % 2.23 % 3 model_cost=0.00%

output_rows=0.00% plan_cost=0.00% rows=0.00%

OperatorCost/SeqScan/2048/real_time_mean

69707 ns 69380 ns 3 model_cost=204.8k

output_rows=2.048k plan_cost=204.8k rows=2.048k

OperatorCost/SeqScan/2048/real_time_median

69583 ns 69238 ns 3 model_cost=204.8k

output_rows=2.048k plan_cost=204.8k rows=2.048k

OperatorCost/SeqScan/2048/real_time_stddev

909 ns 889 ns 3 model_cost=0

output_rows=0 plan_cost=0 rows=0

OperatorCost/SeqScan/2048/real_time_cv

1.30 % 1.28 % 3 model_cost=0.00%

output_rows=0.00% plan_cost=0.00% rows=0.00%

OperatorCost/SeqScan/4096/real_time_mean

154163 ns 152385 ns 3 model_cost=409.6k

output_rows=4.096k plan_cost=409.6k rows=4.096k

OperatorCost/SeqScan/4096/real_time_median

153859 ns 152316 ns 3 model_cost=409.6k

output_rows=4.096k plan_cost=409.6k rows=4.096k

OperatorCost/SeqScan/4096/real_time_stddev

901 ns 563 ns 3 model_cost=0

output_rows=0 plan_cost=0 rows=0

OperatorCost/SeqScan/4096/real_time_cv

0.58 % 0.37 % 3 model_cost=0.00%

output_rows=0.00% plan_cost=0.00% rows=0.00%

```

OperatorCost/SeqScan/8192/real_time_mean
390148 ns      385602 ns      3 model_cost=819.2k
output_rows=8.192k plan_cost=819.2k rows=8.192k
OperatorCost/SeqScan/8192/real_time_median
399998 ns      395298 ns      3 model_cost=819.2k
output_rows=8.192k plan_cost=819.2k rows=8.192k
OperatorCost/SeqScan/8192/real_time_stddev
30561 ns      29527 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/SeqScan/8192/real_time_cv
7.83 %      7.66 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/SeqScan/16384/real_time_mean
924368 ns      914961 ns      3 model_cost=1.6384M
output_rows=16.384k plan_cost=1.6384M rows=16.384k
OperatorCost/SeqScan/16384/real_time_median
931261 ns      921625 ns      3 model_cost=1.6384M
output_rows=16.384k plan_cost=1.6384M rows=16.384k
OperatorCost/SeqScan/16384/real_time_stddev
17627 ns      17412 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/SeqScan/16384/real_time_cv
1.91 %      1.90 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/SeqScan/32768/real_time_mean
1901611 ns      1885331 ns      3 model_cost=3.2768M
output_rows=32.768k plan_cost=3.2768M rows=32.768k
OperatorCost/SeqScan/32768/real_time_median
1933730 ns      1916091 ns      3 model_cost=3.2768M
output_rows=32.768k plan_cost=3.2768M rows=32.768k
OperatorCost/SeqScan/32768/real_time_stddev
90232 ns      87205 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/SeqScan/32768/real_time_cv
4.75 %      4.63 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/SeqScan/65536/real_time_mean
4965998 ns      4909459 ns      3 model_cost=6.5536M
output_rows=65.536k plan_cost=6.5536M rows=65.536k

```



```

OperatorCost/SeqScan/65536/real_time_median
4998141 ns      4946201 ns      3 model_cost=6.5536M
output_rows=65.536k plan_cost=6.5536M rows=65.536k
OperatorCost/SeqScan/65536/real_time_stddev
155421 ns      163668 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/SeqScan/65536/real_time_cv
3.13 %      3.33 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/SeqScan/131072/real_time_mean
12138600 ns      11998804 ns      3 model_cost=13.1072M
output_rows=131.072k plan_cost=13.1072M rows=131.072k
OperatorCost/SeqScan/131072/real_time_median
11162739 ns      11038913 ns      3 model_cost=13.1072M
output_rows=131.072k plan_cost=13.1072M rows=131.072k
OperatorCost/SeqScan/131072/real_time_stddev
1830952 ns      1805064 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/SeqScan/131072/real_time_cv
15.08 %      15.04 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/SeqScan/262144/real_time_mean
25644707 ns      25374424 ns      3 model_cost=26.2144M
output_rows=262.144k plan_cost=26.2144M rows=262.144k
OperatorCost/SeqScan/262144/real_time_median
23244999 ns      23024292 ns      3 model_cost=26.2144M
output_rows=262.144k plan_cost=26.2144M rows=262.144k
OperatorCost/SeqScan/262144/real_time_stddev
4388803 ns      4299232 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/SeqScan/262144/real_time_cv
17.11 %      16.94 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/1024/real_time_mean
70425 ns      70146 ns      3 model_cost=102.4k
output_rows=512 plan_cost=204.8k rows=1.024k
OperatorCost/Filter/1024/real_time_median
70365 ns      70086 ns      3 model_cost=102.4k
output_rows=512 plan_cost=204.8k rows=1.024k

```

```

OperatorCost/Filter/1024/real_time_stddev
535 ns          551 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/1024/real_time_cv
0.76 %          0.79 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/2048/real_time_mean
125482 ns       125054 ns       3 model_cost=204.8k
output_rows=1.024k plan_cost=409.6k rows=2.048k
OperatorCost/Filter/2048/real_time_median
125359 ns       124936 ns       3 model_cost=204.8k
output_rows=1.024k plan_cost=409.6k rows=2.048k
OperatorCost/Filter/2048/real_time_stddev
329 ns          322 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/2048/real_time_cv
0.26 %          0.26 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/4096/real_time_mean
244461 ns       243272 ns       3 model_cost=409.6k
output_rows=2.048k plan_cost=819.2k rows=4.096k
OperatorCost/Filter/4096/real_time_median
244237 ns       242959 ns       3 model_cost=409.6k
output_rows=2.048k plan_cost=819.2k rows=4.096k
OperatorCost/Filter/4096/real_time_stddev
1017 ns         1018 ns         3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/4096/real_time_cv
0.42 %          0.42 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/8192/real_time_mean
479696 ns       477349 ns       3 model_cost=819.2k
output_rows=4.096k plan_cost=1.6384M rows=8.192k
OperatorCost/Filter/8192/real_time_median
477488 ns       475398 ns       3 model_cost=819.2k
output_rows=4.096k plan_cost=1.6384M rows=8.192k
OperatorCost/Filter/8192/real_time_stddev
6388 ns         6050 ns         3 model_cost=0
output_rows=0 plan_cost=0 rows=0

```

```

OperatorCost/Filter/8192/real_time_cv
1.33 %          1.27 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/16384/real_time_mean
993383 ns      988008 ns          3 model_cost=1.6384M
output_rows=8.192k plan_cost=3.2768M rows=16.384k
OperatorCost/Filter/16384/real_time_median
990927 ns      985217 ns          3 model_cost=1.6384M
output_rows=8.192k plan_cost=3.2768M rows=16.384k
OperatorCost/Filter/16384/real_time_stddev
5275 ns        5696 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/16384/real_time_cv
0.53 %          0.58 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/32768/real_time_mean
2178959 ns     2166464 ns          3 model_cost=3.2768M
output_rows=16.384k plan_cost=6.5536M rows=32.768k
OperatorCost/Filter/32768/real_time_median
2129820 ns     2118195 ns          3 model_cost=3.2768M
output_rows=16.384k plan_cost=6.5536M rows=32.768k
OperatorCost/Filter/32768/real_time_stddev
101295 ns      99833 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/32768/real_time_cv
4.65 %          4.61 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/65536/real_time_mean
4436384 ns     4413484 ns          3 model_cost=6.5536M
output_rows=32.768k plan_cost=13.1072M rows=65.536k
OperatorCost/Filter/65536/real_time_median
4473698 ns     4450082 ns          3 model_cost=6.5536M
output_rows=32.768k plan_cost=13.1072M rows=65.536k
OperatorCost/Filter/65536/real_time_stddev
137556 ns      135091 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/65536/real_time_cv
3.10 %          3.06 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%

```

```

OperatorCost/Filter/131072/real_time_mean
9775161 ns      9717069 ns      3 model_cost=13.1072M
output_rows=65.536k plan_cost=26.2144M rows=131.072k
OperatorCost/Filter/131072/real_time_median
9723418 ns      9663916 ns      3 model_cost=13.1072M
output_rows=65.536k plan_cost=26.2144M rows=131.072k
OperatorCost/Filter/131072/real_time_stddev
113812 ns      109969 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/131072/real_time_cv
1.16 %      1.13 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Filter/262144/real_time_mean
20011648 ns      19895345 ns      3 model_cost=26.2144M
output_rows=131.072k plan_cost=52.4288M rows=262.144k
OperatorCost/Filter/262144/real_time_median
20028071 ns      19914759 ns      3 model_cost=26.2144M
output_rows=131.072k plan_cost=52.4288M rows=262.144k
OperatorCost/Filter/262144/real_time_stddev
218024 ns      212289 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Filter/262144/real_time_cv
1.09 %      1.07 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/1024/real_time_mean
94356 ns      94028 ns      3 model_cost=22.528k
output_rows=1.024k plan_cost=124.928k rows=1.024k
OperatorCost/Projection/1024/real_time_median
94592 ns      94279 ns      3 model_cost=22.528k
output_rows=1.024k plan_cost=124.928k rows=1.024k
OperatorCost/Projection/1024/real_time_stddev
520 ns      549 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/1024/real_time_cv
0.55 %      0.58 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/2048/real_time_mean
176410 ns      175840 ns      3 model_cost=45.056k
output_rows=2.048k plan_cost=249.856k rows=2.048k

```

```

OperatorCost/Projection/2048/real_time_median
176334 ns      175777 ns      3 model_cost=45.056k
output_rows=2.048k plan_cost=249.856k rows=2.048k
OperatorCost/Projection/2048/real_time_stddev
553 ns      490 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/2048/real_time_cv
0.31 %      0.28 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/4096/real_time_mean
351209 ns      349742 ns      3 model_cost=90.112k
output_rows=4.096k plan_cost=499.712k rows=4.096k
OperatorCost/Projection/4096/real_time_median
350576 ns      349336 ns      3 model_cost=90.112k
output_rows=4.096k plan_cost=499.712k rows=4.096k
OperatorCost/Projection/4096/real_time_stddev
6628 ns      6489 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/4096/real_time_cv
1.89 %      1.86 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/8192/real_time_mean
688010 ns      684898 ns      3 model_cost=180.224k
output_rows=8.192k plan_cost=0.999424M rows=8.192k
OperatorCost/Projection/8192/real_time_median
687349 ns      684331 ns      3 model_cost=180.224k
output_rows=8.192k plan_cost=0.999424M rows=8.192k
OperatorCost/Projection/8192/real_time_stddev
6078 ns      6140 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/8192/real_time_cv
0.88 %      0.90 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/16384/real_time_mean
1408705 ns      1401811 ns      3 model_cost=360.448k
output_rows=16.384k plan_cost=1.99885M rows=16.384k
OperatorCost/Projection/16384/real_time_median
1405017 ns      1397537 ns      3 model_cost=360.448k
output_rows=16.384k plan_cost=1.99885M rows=16.384k

```

```

OperatorCost/Projection/16384/real_time_stddev
8952 ns          8860 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/16384/real_time_cv
0.64 %          0.63 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/32768/real_time_mean
3063583 ns      3046674 ns          3 model_cost=720.896k
output_rows=32.768k plan_cost=3.9977M rows=32.768k
OperatorCost/Projection/32768/real_time_median
3021017 ns      3006734 ns          3 model_cost=720.896k
output_rows=32.768k plan_cost=3.9977M rows=32.768k
OperatorCost/Projection/32768/real_time_stddev
134698 ns       132627 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/32768/real_time_cv
4.40 %          4.35 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/65536/real_time_mean
6409382 ns      6374643 ns          3 model_cost=1.44179M
output_rows=65.536k plan_cost=7.99539M rows=65.536k
OperatorCost/Projection/65536/real_time_median
6414271 ns      6383091 ns          3 model_cost=1.44179M
output_rows=65.536k plan_cost=7.99539M rows=65.536k
OperatorCost/Projection/65536/real_time_stddev
25434 ns        28186 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/65536/real_time_cv
0.40 %          0.44 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/131072/real_time_mean
13727593 ns     13637236 ns          3 model_cost=2.88358M
output_rows=131.072k plan_cost=15.9908M rows=131.072k
OperatorCost/Projection/131072/real_time_median
13847824 ns     13759305 ns          3 model_cost=2.88358M
output_rows=131.072k plan_cost=15.9908M rows=131.072k
OperatorCost/Projection/131072/real_time_stddev
291770 ns       293141 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0

```

OperatorCost/Projection/131072/real_time_cv
2.13 % 2.15 % 3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Projection/262144/real_time_mean
33247794 ns 33037538 ns 3 model_cost=5.76717M
output_rows=262.144k plan_cost=31.9816M rows=262.144k
OperatorCost/Projection/262144/real_time_median
33275241 ns 33069358 ns 3 model_cost=5.76717M
output_rows=262.144k plan_cost=31.9816M rows=262.144k
OperatorCost/Projection/262144/real_time_stddev
180580 ns 170487 ns 3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Projection/262144/real_time_cv
0.54 % 0.52 % 3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/1024/real_time_mean
114160 ns 113790 ns 3 model_cost=123.904k
output_rows=1.024k plan_cost=226.304k rows=1.024k
OperatorCost/Sort/1024/real_time_median
114326 ns 113955 ns 3 model_cost=123.904k
output_rows=1.024k plan_cost=226.304k rows=1.024k
OperatorCost/Sort/1024/real_time_stddev
589 ns 604 ns 3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/1024/real_time_cv
0.52 % 0.53 % 3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/2048/real_time_mean
266928 ns 266080 ns 3 model_cost=270.336k
output_rows=2.048k plan_cost=475.136k rows=2.048k
OperatorCost/Sort/2048/real_time_median
266698 ns 265906 ns 3 model_cost=270.336k
output_rows=2.048k plan_cost=475.136k rows=2.048k
OperatorCost/Sort/2048/real_time_stddev
743 ns 705 ns 3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/2048/real_time_cv
0.28 % 0.26 % 3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%

```

OperatorCost/Sort/4096/real_time_mean
600007 ns          597438 ns          3 model_cost=585.728k
output_rows=4.096k plan_cost=995.328k rows=4.096k
OperatorCost/Sort/4096/real_time_median
600843 ns          598069 ns          3 model_cost=585.728k
output_rows=4.096k plan_cost=995.328k rows=4.096k
OperatorCost/Sort/4096/real_time_stddev
3889 ns           3608 ns           3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/4096/real_time_cv
0.65 %           0.60 %           3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/8192/real_time_mean
1353285 ns        1347286 ns          3 model_cost=1.26157M
output_rows=8.192k plan_cost=2.08077M rows=8.192k
OperatorCost/Sort/8192/real_time_median
1352224 ns        1346166 ns          3 model_cost=1.26157M
output_rows=8.192k plan_cost=2.08077M rows=8.192k
OperatorCost/Sort/8192/real_time_stddev
3159 ns           3530 ns           3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/8192/real_time_cv
0.23 %           0.26 %           3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/16384/real_time_mean
3330166 ns        3313456 ns          3 model_cost=2.70336M
output_rows=16.384k plan_cost=4.34176M rows=16.384k
OperatorCost/Sort/16384/real_time_median
3280666 ns        3263886 ns          3 model_cost=2.70336M
output_rows=16.384k plan_cost=4.34176M rows=16.384k
OperatorCost/Sort/16384/real_time_stddev
102432 ns         100860 ns          3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/16384/real_time_cv
3.08 %           3.04 %           3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/32768/real_time_mean
8325146 ns        8276866 ns          3 model_cost=5.76717M
output_rows=32.768k plan_cost=9.04397M rows=32.768k

```


OperatorCost/Sort/32768/real_time_median
8328091 ns 8280027 ns 3 model_cost=5.76717M
output_rows=32.768k plan_cost=9.04397M rows=32.768k
OperatorCost/Sort/32768/real_time_stddev
8924 ns 5969 ns 3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/32768/real_time_cv
0.11 % 0.07 % 3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/65536/real_time_mean
22543761 ns 22399862 ns 3 model_cost=12.2552M
output_rows=65.536k plan_cost=18.8088M rows=65.536k
OperatorCost/Sort/65536/real_time_median
21748033 ns 21608460 ns 3 model_cost=12.2552M
output_rows=65.536k plan_cost=18.8088M rows=65.536k
OperatorCost/Sort/65536/real_time_stddev
1392405 ns 1376023 ns 3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/65536/real_time_cv
6.18 % 6.14 % 3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/131072/real_time_mean
63230431 ns 62728449 ns 3 model_cost=25.9523M
output_rows=131.072k plan_cost=39.0595M rows=131.072k
OperatorCost/Sort/131072/real_time_median
62867420 ns 62387321 ns 3 model_cost=25.9523M
output_rows=131.072k plan_cost=39.0595M rows=131.072k
OperatorCost/Sort/131072/real_time_stddev
1185317 ns 1168169 ns 3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/131072/real_time_cv
1.87 % 1.86 % 3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Sort/262144/real_time_mean
176649519 ns 175484283 ns 3 model_cost=54.7881M
output_rows=262.144k plan_cost=81.0025M rows=262.144k
OperatorCost/Sort/262144/real_time_median
175323726 ns 174144845 ns 3 model_cost=54.7881M
output_rows=262.144k plan_cost=81.0025M rows=262.144k

```

OperatorCost/Sort/262144/real_time_stddev
6750024 ns      6601525 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Sort/262144/real_time_cv
3.82 %      3.76 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/1024/real_time_mean
274703 ns      273781 ns      3 model_cost=522.24k
output_rows=1.024k plan_cost=624.64k rows=1.024k
OperatorCost/Aggregation/1024/real_time_median
275378 ns      274420 ns      3 model_cost=522.24k
output_rows=1.024k plan_cost=624.64k rows=1.024k
OperatorCost/Aggregation/1024/real_time_stddev
1834 ns      1818 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/1024/real_time_cv
0.67 %      0.66 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/2048/real_time_mean
547245 ns      544909 ns      3 model_cost=1.04448M
output_rows=2.048k plan_cost=1.24928M rows=2.048k
OperatorCost/Aggregation/2048/real_time_median
542243 ns      539784 ns      3 model_cost=1.04448M
output_rows=2.048k plan_cost=1.24928M rows=2.048k
OperatorCost/Aggregation/2048/real_time_stddev
15295 ns      15170 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/2048/real_time_cv
2.79 %      2.78 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/4096/real_time_mean
1141244 ns      1135478 ns      3 model_cost=2.08896M
output_rows=4.096k plan_cost=2.49856M rows=4.096k
OperatorCost/Aggregation/4096/real_time_median
1143881 ns      1138252 ns      3 model_cost=2.08896M
output_rows=4.096k plan_cost=2.49856M rows=4.096k
OperatorCost/Aggregation/4096/real_time_stddev
15733 ns      15158 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0

```

```

OperatorCost/Aggregation/4096/real_time_cv
1.38 %          1.33 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/8192/real_time_mean
2426264 ns      2414370 ns      3 model_cost=4.17792M
output_rows=8.192k plan_cost=4.99712M rows=8.192k
OperatorCost/Aggregation/8192/real_time_median
2422686 ns      2411175 ns      3 model_cost=4.17792M
output_rows=8.192k plan_cost=4.99712M rows=8.192k
OperatorCost/Aggregation/8192/real_time_stddev
43691 ns        43834 ns        3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/8192/real_time_cv
1.80 %          1.82 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/16384/real_time_mean
5812245 ns      5776502 ns      3 model_cost=8.35584M
output_rows=16.384k plan_cost=9.99424M rows=16.384k
OperatorCost/Aggregation/16384/real_time_median
5826207 ns      5793681 ns      3 model_cost=8.35584M
output_rows=16.384k plan_cost=9.99424M rows=16.384k
OperatorCost/Aggregation/16384/real_time_stddev
108021 ns       105107 ns       3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/16384/real_time_cv
1.86 %          1.82 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/32768/real_time_mean
16768921 ns     16655176 ns     3 model_cost=16.7117M
output_rows=32.768k plan_cost=19.9885M rows=32.768k
OperatorCost/Aggregation/32768/real_time_median
16630918 ns     16514790 ns     3 model_cost=16.7117M
output_rows=32.768k plan_cost=19.9885M rows=32.768k
OperatorCost/Aggregation/32768/real_time_stddev
241465 ns       245756 ns       3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/32768/real_time_cv
1.44 %          1.48 %          3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%

```

```

OperatorCost/Aggregation/65536/real_time_mean
52625470 ns      52271134 ns      3 model_cost=33.4234M
output_rows=65.536k plan_cost=39.977M rows=65.536k
OperatorCost/Aggregation/65536/real_time_median
52315635 ns      51975856 ns      3 model_cost=33.4234M
output_rows=65.536k plan_cost=39.977M rows=65.536k
OperatorCost/Aggregation/65536/real_time_stddev
2766613 ns      2683321 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/65536/real_time_cv
5.26 %      5.13 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/131072/real_time_mean
129696984 ns      128861029 ns      3 model_cost=66.8467M
output_rows=131.072k plan_cost=79.9539M rows=131.072k
OperatorCost/Aggregation/131072/real_time_median
128465716 ns      127653219 ns      3 model_cost=66.8467M
output_rows=131.072k plan_cost=79.9539M rows=131.072k
OperatorCost/Aggregation/131072/real_time_stddev
3404042 ns      3352711 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/131072/real_time_cv
2.62 %      2.60 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/Aggregation/262144/real_time_mean
300294351 ns      298535917 ns      3 model_cost=133.693M
output_rows=262.144k plan_cost=159.908M rows=262.144k
OperatorCost/Aggregation/262144/real_time_median
298677499 ns      297021558 ns      3 model_cost=133.693M
output_rows=262.144k plan_cost=159.908M rows=262.144k
OperatorCost/Aggregation/262144/real_time_stddev
11210695 ns      10988337 ns      3 model_cost=0
output_rows=0 plan_cost=0 rows=0
OperatorCost/Aggregation/262144/real_time_cv
3.73 %      3.68 %      3 model_cost=0.00%
output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCost/HashJoin/1024/1024/real_time_mean
155098 ns      154565 ns      3 lhs_rows=1.024k
model_cost=404.48k output_rows=1.024k plan_cost=609.28k
rhs_rows=1.024k

```

OperatorCost/HashJoin/1024/1024/real_time_median
155342 ns 154745 ns 3 lhs_rows=1.024k
model_cost=404.48k output_rows=1.024k plan_cost=609.28k
rhs_rows=1.024k
OperatorCost/HashJoin/1024/1024/real_time_stddev
1119 ns 1070 ns 3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/HashJoin/1024/1024/real_time_cv
0.72 % 0.69 % 3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/HashJoin/2048/2048/real_time_mean
301359 ns 300041 ns 3 lhs_rows=2.048k
model_cost=808.96k output_rows=2.048k plan_cost=1.21856M
rhs_rows=2.048k
OperatorCost/HashJoin/2048/2048/real_time_median
302359 ns 301088 ns 3 lhs_rows=2.048k
model_cost=808.96k output_rows=2.048k plan_cost=1.21856M
rhs_rows=2.048k
OperatorCost/HashJoin/2048/2048/real_time_stddev
2615 ns 2492 ns 3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/HashJoin/2048/2048/real_time_cv
0.87 % 0.83 % 3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/HashJoin/4096/4096/real_time_mean
599260 ns 596096 ns 3 lhs_rows=4.096k
model_cost=1.61792M output_rows=4.096k plan_cost=2.43712M
rhs_rows=4.096k
OperatorCost/HashJoin/4096/4096/real_time_median
599183 ns 596316 ns 3 lhs_rows=4.096k
model_cost=1.61792M output_rows=4.096k plan_cost=2.43712M
rhs_rows=4.096k
OperatorCost/HashJoin/4096/4096/real_time_stddev
2586 ns 2808 ns 3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/HashJoin/4096/4096/real_time_cv
0.43 % 0.47 % 3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/HashJoin/8192/8192/real_time_mean
1274549 ns 1266452 ns 3 lhs_rows=8.192k
model_cost=3.23584M output_rows=8.192k plan_cost=4.87424M

```

rhs_rows=8.192k
OperatorCost/HashJoin/8192/8192/real_time_median
1234395 ns      1226034 ns      3 lhs_rows=8.192k
model_cost=3.23584M output_rows=8.192k plan_cost=4.87424M
rhs_rows=8.192k
OperatorCost/HashJoin/8192/8192/real_time_stddev
78831 ns      77036 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/HashJoin/8192/8192/real_time_cv
6.18 %      6.08 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/HashJoin/16384/16384/real_time_mean
2941091 ns      2919158 ns      3 lhs_rows=16.384k
model_cost=6.47168M output_rows=16.384k plan_cost=9.74848M
rhs_rows=16.384k
OperatorCost/HashJoin/16384/16384/real_time_median
2911038 ns      2889304 ns      3 lhs_rows=16.384k
model_cost=6.47168M output_rows=16.384k plan_cost=9.74848M
rhs_rows=16.384k
OperatorCost/HashJoin/16384/16384/real_time_stddev
67545 ns      67104 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/HashJoin/16384/16384/real_time_cv
2.30 %      2.30 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/HashJoin/32768/32768/real_time_mean
6940540 ns      6889217 ns      3 lhs_rows=32.768k
model_cost=12.9434M output_rows=32.768k plan_cost=19.497M
rhs_rows=32.768k
OperatorCost/HashJoin/32768/32768/real_time_median
6908295 ns      6859205 ns      3 lhs_rows=32.768k
model_cost=12.9434M output_rows=32.768k plan_cost=19.497M
rhs_rows=32.768k
OperatorCost/HashJoin/32768/32768/real_time_stddev
70111 ns      72151 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/HashJoin/32768/32768/real_time_cv
1.01 %      1.05 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/HashJoin/65536/65536/real_time_mean
15832571 ns      15709972 ns      3 lhs_rows=65.536k

```

```

model_cost=25.8867M output_rows=65.536k plan_cost=38.9939M
rhs_rows=65.536k
OperatorCost/HashJoin/65536/65536/real_time_median
15705191 ns      15604051 ns      3 lhs_rows=65.536k
model_cost=25.8867M output_rows=65.536k plan_cost=38.9939M
rhs_rows=65.536k
OperatorCost/HashJoin/65536/65536/real_time_stddev
240004 ns      235698 ns      3 lhs_rows=0
model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/HashJoin/65536/65536/real_time_cv
1.52 %      1.50 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopJoin/64/64/real_time_mean
341299 ns      339396 ns      3 lhs_rows=64
model_cost=286.72k output_rows=64 plan_cost=299.52k rhs_rows=64
OperatorCost/NestedLoopJoin/64/64/real_time_median
340857 ns      338876 ns      3 lhs_rows=64
model_cost=286.72k output_rows=64 plan_cost=299.52k rhs_rows=64
OperatorCost/NestedLoopJoin/64/64/real_time_stddev
7369 ns      7193 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopJoin/64/64/real_time_cv
2.16 %      2.12 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopJoin/128/128/real_time_mean
1072887 ns      1067776 ns      3 lhs_rows=128
model_cost=1.14688M output_rows=128 plan_cost=1.17248M rhs_rows=128
OperatorCost/NestedLoopJoin/128/128/real_time_median
1069796 ns      1064744 ns      3 lhs_rows=128
model_cost=1.14688M output_rows=128 plan_cost=1.17248M rhs_rows=128
OperatorCost/NestedLoopJoin/128/128/real_time_stddev
8259 ns      8226 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopJoin/128/128/real_time_cv
0.77 %      0.77 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopJoin/256/256/real_time_mean
3943250 ns      3924191 ns      3 lhs_rows=256
model_cost=4.58752M output_rows=256 plan_cost=4.63872M rhs_rows=256

```

```

OperatorCost/NestedLoopJoin/256/256/real_time_median
3886867 ns      3870416 ns      3 lhs_rows=256
model_cost=4.58752M output_rows=256 plan_cost=4.63872M rhs_rows=256
OperatorCost/NestedLoopJoin/256/256/real_time_stddev
115332 ns      111745 ns      3 lhs_rows=0
model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopJoin/256/256/real_time_cv
2.92 %      2.85 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopJoin/512/512/real_time_mean
14965494 ns      14909916 ns      3 lhs_rows=512
model_cost=18.3501M output_rows=512 plan_cost=18.4525M rhs_rows=512
OperatorCost/NestedLoopJoin/512/512/real_time_median
15020643 ns      14962434 ns      3 lhs_rows=512
model_cost=18.3501M output_rows=512 plan_cost=18.4525M rhs_rows=512
OperatorCost/NestedLoopJoin/512/512/real_time_stddev
136569 ns      133694 ns      3 lhs_rows=0
model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopJoin/512/512/real_time_cv
0.91 %      0.90 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopJoin/1024/1024/real_time_mean
57691018 ns      57436935 ns      3 lhs_rows=1.024k
model_cost=73.4003M output_rows=1.024k plan_cost=73.6051M
rhs_rows=1.024k
OperatorCost/NestedLoopJoin/1024/1024/real_time_median
57607890 ns      57430034 ns      3 lhs_rows=1.024k
model_cost=73.4003M output_rows=1.024k plan_cost=73.6051M
rhs_rows=1.024k
OperatorCost/NestedLoopJoin/1024/1024/real_time_stddev
451356 ns      317271 ns      3 lhs_rows=0
model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopJoin/1024/1024/real_time_cv
0.78 %      0.55 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopJoin/2048/2048/real_time_mean
229988242 ns      229099036 ns      3 lhs_rows=2.048k
model_cost=293.601M output_rows=2.048k plan_cost=294.011M
rhs_rows=2.048k

```



```

OperatorCost/NestedLoopJoin/2048/2048/real_time_median
229828521 ns      229065432 ns      3 lhs_rows=2.048k
model_cost=293.601M output_rows=2.048k plan_cost=294.011M
rhs_rows=2.048k
OperatorCost/NestedLoopJoin/2048/2048/real_time_stddev
2792600 ns      2747616 ns      3 lhs_rows=0
model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopJoin/2048/2048/real_time_cv
1.21 %      1.20 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopCrossJoin/32/32/real_time_mean
130942 ns      129796 ns      3 lhs_rows=32
model_cost=106.496k output_rows=1.024k plan_cost=112.896k
rhs_rows=32
OperatorCost/NestedLoopCrossJoin/32/32/real_time_median
131308 ns      130105 ns      3 lhs_rows=32
model_cost=106.496k output_rows=1.024k plan_cost=112.896k
rhs_rows=32
OperatorCost/NestedLoopCrossJoin/32/32/real_time_stddev
1935 ns      1924 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopCrossJoin/32/32/real_time_cv
1.48 %      1.48 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopCrossJoin/64/64/real_time_mean
288639 ns      286761 ns      3 lhs_rows=64
model_cost=425.984k output_rows=4.096k plan_cost=438.784k
rhs_rows=64
OperatorCost/NestedLoopCrossJoin/64/64/real_time_median
288545 ns      286708 ns      3 lhs_rows=64
model_cost=425.984k output_rows=4.096k plan_cost=438.784k
rhs_rows=64
OperatorCost/NestedLoopCrossJoin/64/64/real_time_stddev
3202 ns      3171 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopCrossJoin/64/64/real_time_cv
1.11 %      1.11 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopCrossJoin/128/128/real_time_mean
899354 ns      893066 ns      3 lhs_rows=128
model_cost=1.70394M output_rows=16.384k plan_cost=1.72954M

```

```

rhs_rows=128
OperatorCost/NestedLoopCrossJoin/128/128/real_time_median
881125 ns      874885 ns      3 lhs_rows=128
model_cost=1.70394M output_rows=16.384k plan_cost=1.72954M
rhs_rows=128
OperatorCost/NestedLoopCrossJoin/128/128/real_time_stddev
52481 ns      51442 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopCrossJoin/128/128/real_time_cv
5.84 %      5.76 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopCrossJoin/256/256/real_time_mean
5036885 ns      4983636 ns      3 lhs_rows=256
model_cost=6.81574M output_rows=65.536k plan_cost=6.86694M
rhs_rows=256
OperatorCost/NestedLoopCrossJoin/256/256/real_time_median
5043142 ns      4984503 ns      3 lhs_rows=256
model_cost=6.81574M output_rows=65.536k plan_cost=6.86694M
rhs_rows=256
OperatorCost/NestedLoopCrossJoin/256/256/real_time_stddev
47815 ns      44463 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopCrossJoin/256/256/real_time_cv
0.95 %      0.89 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopCrossJoin/512/512/real_time_mean
25155229 ns      24914236 ns      3 lhs_rows=512
model_cost=27.263M output_rows=262.144k plan_cost=27.3654M
rhs_rows=512
OperatorCost/NestedLoopCrossJoin/512/512/real_time_median
25126405 ns      24885776 ns      3 lhs_rows=512
model_cost=27.263M output_rows=262.144k plan_cost=27.3654M
rhs_rows=512
OperatorCost/NestedLoopCrossJoin/512/512/real_time_stddev
71506 ns      52169 ns      3 lhs_rows=0 model_cost=0
output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopCrossJoin/512/512/real_time_cv
0.28 %      0.21 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%
OperatorCost/NestedLoopCrossJoin/1024/1024/real_time_mean
125736836 ns      124681210 ns      3 lhs_rows=1.024k

```

```

model_cost=109.052M output_rows=1.04858M plan_cost=109.257M
rhs_rows=1.024k
OperatorCost/NestedLoopCrossJoin/1024/1024/real_time_median
124916767 ns      123921643 ns      3 lhs_rows=1.024k
model_cost=109.052M output_rows=1.04858M plan_cost=109.257M
rhs_rows=1.024k
OperatorCost/NestedLoopCrossJoin/1024/1024/real_time_stddev
2950666 ns      2904895 ns      3 lhs_rows=0
model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCost/NestedLoopCrossJoin/1024/1024/real_time_cv
2.35 %      2.33 %      3 lhs_rows=0.00%
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%

```

```

[5]: !cd .. && timeout 300s ./build-release/bin/benchmarks
      ↪--benchmark_filter='^OperatorCostMatched/' --benchmark_repetitions=3
      ↪--benchmark_display_aggregates_only=true --benchmark_out=research/
      ↪benchmark-results/operator-cost-matched.json --benchmark_out_format=json

```

```

2026-06-02T01:20:14+03:00
Running ./build-release/bin/benchmarks
Run on (8 X 4200 MHz CPU s)
CPU Caches:
  L1 Data 48 KiB (x4)
  L1 Instruction 32 KiB (x4)
  L2 Unified 1280 KiB (x4)
  L3 Unified 8192 KiB (x1)
Load Average: 1.60, 1.07, 0.73
***WARNING*** CPU scaling is enabled, the benchmark real time measurements may
be noisy and will incur extra overhead.

```

```

-----
Benchmark
Time          CPU    Iterations UserCounters...
-----
OperatorCostMatched/target_cost:640000/SeqScan/6400/real_time_mean
      320159 ns      317346 ns      3
model_cost=640k output_rows=6.4k plan_cost=640k rows=6.4k
OperatorCostMatched/target_cost:640000/SeqScan/6400/real_time_median
      319599 ns      316785 ns      3
model_cost=640k output_rows=6.4k plan_cost=640k rows=6.4k
OperatorCostMatched/target_cost:640000/SeqScan/6400/real_time_stddev
      1160 ns      1115 ns      3
model_cost=0 output_rows=0 plan_cost=0 rows=0

```

OperatorCostMatched/target_cost:640000/SeqScan/6400/real_time_cv
0.36 % 0.35 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:640000/Filter/6400/real_time_mean
409248 ns 407255 ns 3
model_cost=640k output_rows=3.328k plan_cost=1.28M rows=6.4k
OperatorCostMatched/target_cost:640000/Filter/6400/real_time_median
399719 ns 397882 ns 3
model_cost=640k output_rows=3.328k plan_cost=1.28M rows=6.4k
OperatorCostMatched/target_cost:640000/Filter/6400/real_time_stddev
18699 ns 18531 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:640000/Filter/6400/real_time_cv
4.57 % 4.55 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:640000/Projection/29091/real_time_mean
3097292 ns 3075629 ns 3
model_cost=640.002k output_rows=29.091k plan_cost=3.5491M
rows=29.091k
OperatorCostMatched/target_cost:640000/Projection/29091/real_time_med
an 3113549 ns 3091459 ns
3 model_cost=640.002k output_rows=29.091k plan_cost=3.5491M
rows=29.091k
OperatorCostMatched/target_cost:640000/Projection/29091/real_time_std
ev 46014 ns 45803 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:640000/Projection/29091/real_time_cv
1.49 % 1.49 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:640000/Sort/4476/real_time_mean
714273 ns 707808 ns 3
model_cost=640.068k output_rows=4.476k plan_cost=1.08767M
rows=4.476k
OperatorCostMatched/target_cost:640000/Sort/4476/real_time_median
704145 ns 697639 ns 3
model_cost=640.068k output_rows=4.476k plan_cost=1.08767M
rows=4.476k
OperatorCostMatched/target_cost:640000/Sort/4476/real_time_stddev
21201 ns 20423 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0

OperatorCostMatched/target_cost:640000/Sort/4476/real_time_cv

2.97 %	2.89 %	3
--------	--------	---

model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%

OperatorCostMatched/target_cost:640000/Aggregation/1255/real_time_mean

353749 ns	351405 ns	3
-----------	-----------	---

model_cost=640.05k output_rows=1.255k plan_cost=765.55k rows=1.255k

OperatorCostMatched/target_cost:640000/Aggregation/1255/real_time_median

352784 ns	350809 ns	
-----------	-----------	--

3 model_cost=640.05k output_rows=1.255k plan_cost=765.55k rows=1.255k

OperatorCostMatched/target_cost:640000/Aggregation/1255/real_time_stddev

1705 ns	2169 ns	
---------	---------	--

3 model_cost=0 output_rows=0 plan_cost=0 rows=0

OperatorCostMatched/target_cost:640000/Aggregation/1255/real_time_cv

0.48 %	0.62 %	3
--------	--------	---

model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%

OperatorCostMatched/target_cost:640000/HashJoin/1620/1620/real_time_mean

251418 ns	250042 ns	
-----------	-----------	--

3 lhs_rows=1.62k model_cost=639.9k output_rows=1.62k plan_cost=963.9k rhs_rows=1.62k

OperatorCostMatched/target_cost:640000/HashJoin/1620/1620/real_time_median

250657 ns	249485 ns	
-----------	-----------	--

3 lhs_rows=1.62k model_cost=639.9k output_rows=1.62k plan_cost=963.9k rhs_rows=1.62k

OperatorCostMatched/target_cost:640000/HashJoin/1620/1620/real_time_stddev

1990 ns	1973 ns	
---------	---------	--

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0

OperatorCostMatched/target_cost:640000/HashJoin/1620/1620/real_time_cv

0.79 %	0.79 %	3
--------	--------	---

lhs_rows=0.00% model_cost=0.00% output_rows=0.00% plan_cost=0.00% rhs_rows=0.00%

OperatorCostMatched/target_cost:640000/NestedLoopJoin/96/96/real_time_mean

676351 ns	670795 ns	
-----------	-----------	--

3 lhs_rows=96 model_cost=645.12k output_rows=96 plan_cost=664.32k rhs_rows=96

OperatorCostMatched/target_cost:640000/NestedLoopJoin/96/96/real_time_median

675430 ns	670332 ns	
-----------	-----------	--

3 lhs_rows=96 model_cost=645.12k output_rows=96 plan_cost=664.32k

```

rhs_rows=96
OperatorCostMatched/target_cost:640000/NestedLoopJoin/96/96/real_time_
stddev                2046 ns                2832 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:640000/NestedLoopJoin/96/96/real_time_
cv                    0.30 %                0.42 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:640000/NestedLoopCrossJoin/78/78/real_
time_mean             398744 ns            394565 ns

3 lhs_rows=78 model_cost=632.736k output_rows=6.084k
plan_cost=648.336k rhs_rows=78
OperatorCostMatched/target_cost:640000/NestedLoopCrossJoin/78/78/real_
time_median           399361 ns            393912 ns

3 lhs_rows=78 model_cost=632.736k output_rows=6.084k
plan_cost=648.336k rhs_rows=78
OperatorCostMatched/target_cost:640000/NestedLoopCrossJoin/78/78/real_
time_stddev           3592 ns              3590 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:640000/NestedLoopCrossJoin/78/78/real_
time_cv               0.90 %              0.91 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:1000000/SeqScan/10000/real_time_mean
                    352346 ns            350260 ns                3
model_cost=1M output_rows=10k plan_cost=1M rows=10k
OperatorCostMatched/target_cost:1000000/SeqScan/10000/real_time_median
                    349456 ns            347781 ns                3
model_cost=1M output_rows=10k plan_cost=1M rows=10k
OperatorCostMatched/target_cost:1000000/SeqScan/10000/real_time_stddev
                    6767 ns              6945 ns                3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:1000000/SeqScan/10000/real_time_cv
                    1.92 %              1.98 %                3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:1000000/Filter/10000/real_time_mean
                    664349 ns            660903 ns                3
model_cost=1M output_rows=5.12k plan_cost=2M rows=10k

```

OperatorCostMatched/target_cost:1000000/Filter/10000/real_time_median
655750 ns 652271 ns 3
model_cost=1M output_rows=5.12k plan_cost=2M rows=10k
OperatorCostMatched/target_cost:1000000/Filter/10000/real_time_stddev
24445 ns 24189 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:1000000/Filter/10000/real_time_cv
3.68 % 3.66 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:1000000/Projection/45455/real_time_mean
n 5560017 ns 5518016 ns
3 model_cost=1.00001M output_rows=45.455k plan_cost=5.54551M
rows=45.455k
OperatorCostMatched/target_cost:1000000/Projection/45455/real_time_median
5563214 ns 5519348 ns
3 model_cost=1.00001M output_rows=45.455k plan_cost=5.54551M
rows=45.455k
OperatorCostMatched/target_cost:1000000/Projection/45455/real_time_stddev
17569 ns 19382 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:1000000/Projection/45455/real_time_cv
0.32 % 0.35 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:1000000/Sort/6993/real_time_mean
1181814 ns 1176185 ns 3
model_cost=0.999999M output_rows=6.993k plan_cost=1.6993M
rows=6.993k
OperatorCostMatched/target_cost:1000000/Sort/6993/real_time_median
1183289 ns 1177677 ns 3
model_cost=0.999999M output_rows=6.993k plan_cost=1.6993M
rows=6.993k
OperatorCostMatched/target_cost:1000000/Sort/6993/real_time_stddev
7241 ns 6718 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:1000000/Sort/6993/real_time_cv
0.61 % 0.57 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%

```

OperatorCostMatched/target_cost:1000000/Aggregation/1961/real_time_mean
n                    540896 ns          538945 ns
3 model_cost=1.00011M output_rows=1.961k plan_cost=1.19621M
rows=1.961k
OperatorCostMatched/target_cost:1000000/Aggregation/1961/real_time_median
                    539330 ns          537405 ns
3 model_cost=1.00011M output_rows=1.961k plan_cost=1.19621M
rows=1.961k
OperatorCostMatched/target_cost:1000000/Aggregation/1961/real_time_stddev
                    3827 ns           3777 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:1000000/Aggregation/1961/real_time_cv
                    0.71 %           0.70 %           3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:1000000/HashJoin/2532/2532/real_time_mean
                    408221 ns          406089 ns
3 lhs_rows=2.532k model_cost=1.00014M output_rows=2.532k
plan_cost=1.50654M rhs_rows=2.532k
OperatorCostMatched/target_cost:1000000/HashJoin/2532/2532/real_time_median
                    413762 ns          411413 ns
3 lhs_rows=2.532k model_cost=1.00014M output_rows=2.532k
plan_cost=1.50654M rhs_rows=2.532k
OperatorCostMatched/target_cost:1000000/HashJoin/2532/2532/real_time_stddev
                    10162 ns           9830 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:1000000/HashJoin/2532/2532/real_time_cv
                    2.49 %           2.42 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:1000000/NestedLoopJoin/120/120/real_time_mean
                    1006221 ns          997401 ns
3 lhs_rows=120 model_cost=1.008M output_rows=120 plan_cost=1.032M
rhs_rows=120
OperatorCostMatched/target_cost:1000000/NestedLoopJoin/120/120/real_time_median
                    1002051 ns          996991 ns
3 lhs_rows=120 model_cost=1.008M output_rows=120 plan_cost=1.032M
rhs_rows=120

```



```

OperatorCostMatched/target_cost:1000000/NestedLoopJoin/120/120/real_time_mean
me_stddev          7815 ns          1517 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:1000000/NestedLoopJoin/120/120/real_time_cv
me_cv              0.78 %          0.15 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:1000000/NestedLoopCrossJoin/98/98/real_time_mean
_time_mean        600207 ns        596002 ns
3 lhs_rows=98 model_cost=998.816k output_rows=9.604k
plan_cost=1.01842M rhs_rows=98
OperatorCostMatched/target_cost:1000000/NestedLoopCrossJoin/98/98/real_time_median
_time_median      600347 ns        596164 ns
3 lhs_rows=98 model_cost=998.816k output_rows=9.604k
plan_cost=1.01842M rhs_rows=98
OperatorCostMatched/target_cost:1000000/NestedLoopCrossJoin/98/98/real_time_stddev
_time_stddev      1109 ns         1109 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:1000000/NestedLoopCrossJoin/98/98/real_time_cv
_time_cv          0.18 %          0.19 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:3240000/SeqScan/32400/real_time_mean
1284652 ns        1276040 ns        3
model_cost=3.24M output_rows=32.4k plan_cost=3.24M rows=32.4k
OperatorCostMatched/target_cost:3240000/SeqScan/32400/real_time_median
1257372 ns        1248532 ns        3
model_cost=3.24M output_rows=32.4k plan_cost=3.24M rows=32.4k
OperatorCostMatched/target_cost:3240000/SeqScan/32400/real_time_stddev
75695 ns          74938 ns          3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:3240000/SeqScan/32400/real_time_cv
5.89 %            5.87 %            3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:3240000/Filter/32400/real_time_mean
2176167 ns        2165969 ns        3
model_cost=3.24M output_rows=16.384k plan_cost=6.48M rows=32.4k

```

OperatorCostMatched/target_cost:3240000/Filter/32400/real_time_median
2159127 ns 2150028 ns 3
model_cost=3.24M output_rows=16.384k plan_cost=6.48M rows=32.4k
OperatorCostMatched/target_cost:3240000/Filter/32400/real_time_stddev
45990 ns 44955 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:3240000/Filter/32400/real_time_cv
2.11 % 2.08 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:3240000/Projection/147273/real_time_me
an 20058836 ns 19892307 ns
3 model_cost=3.24001M output_rows=147.273k plan_cost=17.9673M
rows=147.273k
OperatorCostMatched/target_cost:3240000/Projection/147273/real_time_me
dian 19959603 ns 19781221 ns
3 model_cost=3.24001M output_rows=147.273k plan_cost=17.9673M
rows=147.273k
OperatorCostMatched/target_cost:3240000/Projection/147273/real_time_st
ddev 272114 ns 258775 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:3240000/Projection/147273/real_time_cv
1.36 % 1.30 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:3240000/Sort/19636/real_time_mean
4184701 ns 4161788 ns 3
model_cost=3.23994M output_rows=19.636k plan_cost=5.20354M
rows=19.636k
OperatorCostMatched/target_cost:3240000/Sort/19636/real_time_median
4182936 ns 4160248 ns 3
model_cost=3.23994M output_rows=19.636k plan_cost=5.20354M
rows=19.636k
OperatorCostMatched/target_cost:3240000/Sort/19636/real_time_stddev
21162 ns 17617 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:3240000/Sort/19636/real_time_cv
0.51 % 0.42 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%

```

OperatorCostMatched/target_cost:3240000/Aggregation/6353/real_time_mean
n                1881536 ns        1872941 ns
3 model_cost=3.24003M output_rows=6.353k plan_cost=3.87533M
rows=6.353k
OperatorCostMatched/target_cost:3240000/Aggregation/6353/real_time_median
ian              1876677 ns        1867856 ns
3 model_cost=3.24003M output_rows=6.353k plan_cost=3.87533M
rows=6.353k
OperatorCostMatched/target_cost:3240000/Aggregation/6353/real_time_stddev
dev              19364 ns          18755 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:3240000/Aggregation/6353/real_time_cv
1.03 %           1.00 %           3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:3240000/HashJoin/8203/8203/real_time_mean
ean              1306558 ns        1298432 ns
3 lhs_rows=8.203k model_cost=3.24018M output_rows=8.203k
plan_cost=4.88078M rhs_rows=8.203k
OperatorCostMatched/target_cost:3240000/HashJoin/8203/8203/real_time_median
edian           1306181 ns        1297919 ns
3 lhs_rows=8.203k model_cost=3.24018M output_rows=8.203k
plan_cost=4.88078M rhs_rows=8.203k
OperatorCostMatched/target_cost:3240000/HashJoin/8203/8203/real_time_stddev
tddev            1248 ns          1251 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:3240000/HashJoin/8203/8203/real_time_cv
v                0.10 %           0.10 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:3240000/NestedLoopJoin/215/215/real_time_mean
me_mean          2883331 ns        2862880 ns
3 lhs_rows=215 model_cost=3.23575M output_rows=215
plan_cost=3.27875M rhs_rows=215
OperatorCostMatched/target_cost:3240000/NestedLoopJoin/215/215/real_time_median
me_median         2883138 ns        2870337 ns
3 lhs_rows=215 model_cost=3.23575M output_rows=215
plan_cost=3.27875M rhs_rows=215

```

```

OperatorCostMatched/target_cost:3240000/NestedLoopJoin/215/215/real_time_mean
me_stddev          25044 ns          36841 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:3240000/NestedLoopJoin/215/215/real_time_cv
me_cv              0.87 %          1.29 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:3240000/NestedLoopCrossJoin/177/177/real_time_mean
al_time_mean       1967699 ns       1948450 ns
3 lhs_rows=177 model_cost=3.25822M output_rows=31.329k
plan_cost=3.29362M rhs_rows=177
OperatorCostMatched/target_cost:3240000/NestedLoopCrossJoin/177/177/real_time_median
al_time_median      1961065 ns       1941733 ns
3 lhs_rows=177 model_cost=3.25822M output_rows=31.329k
plan_cost=3.29362M rhs_rows=177
OperatorCostMatched/target_cost:3240000/NestedLoopCrossJoin/177/177/real_time_stddev
al_time_stddev      26641 ns        26322 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:3240000/NestedLoopCrossJoin/177/177/real_time_cv
al_time_cv          1.35 %          1.35 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:6760000/SeqScan/67600/real_time_mean
3804148 ns          3769127 ns          3
model_cost=6.76M output_rows=67.6k plan_cost=6.76M rows=67.6k
OperatorCostMatched/target_cost:6760000/SeqScan/67600/real_time_median
3743306 ns          3710163 ns          3
model_cost=6.76M output_rows=67.6k plan_cost=6.76M rows=67.6k
OperatorCostMatched/target_cost:6760000/SeqScan/67600/real_time_stddev
161694 ns           156528 ns          3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:6760000/SeqScan/67600/real_time_cv
4.25 %              4.15 %          3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:6760000/Filter/67600/real_time_mean
4921571 ns          4893957 ns          3
model_cost=6.76M output_rows=33.808k plan_cost=13.52M rows=67.6k

```

OperatorCostMatched/target_cost:6760000/Filter/67600/real_time_median
4900259 ns 4872979 ns 3
model_cost=6.76M output_rows=33.808k plan_cost=13.52M rows=67.6k
OperatorCostMatched/target_cost:6760000/Filter/67600/real_time_stddev
133893 ns 133878 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:6760000/Filter/67600/real_time_cv
2.72 % 2.74 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:6760000/Projection/307273/real_time_me
an 41998731 ns 41673692 ns
3 model_cost=6.76001M output_rows=307.273k plan_cost=37.4873M
rows=307.273k
OperatorCostMatched/target_cost:6760000/Projection/307273/real_time_me
dian 41714916 ns 41392995 ns
3 model_cost=6.76001M output_rows=307.273k plan_cost=37.4873M
rows=307.273k
OperatorCostMatched/target_cost:6760000/Projection/307273/real_time_st
ddev 535414 ns 528570 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:6760000/Projection/307273/real_time_cv
1.27 % 1.27 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:6760000/Sort/38409/real_time_mean
10271929 ns 10215490 ns 3
model_cost=6.75998M output_rows=38.409k plan_cost=10.6009M
rows=38.409k
OperatorCostMatched/target_cost:6760000/Sort/38409/real_time_median
10293266 ns 10238813 ns 3
model_cost=6.75998M output_rows=38.409k plan_cost=10.6009M
rows=38.409k
OperatorCostMatched/target_cost:6760000/Sort/38409/real_time_stddev
86829 ns 86031 ns 3
model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:6760000/Sort/38409/real_time_cv
0.85 % 0.84 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%

```

OperatorCostMatched/target_cost:6760000/Aggregation/13255/real_time_me
an                4438157 ns        4412963 ns

3 model_cost=6.76005M output_rows=13.255k plan_cost=8.08555M
rows=13.255k
OperatorCostMatched/target_cost:6760000/Aggregation/13255/real_time_me
dian              4443256 ns        4415243 ns

3 model_cost=6.76005M output_rows=13.255k plan_cost=8.08555M
rows=13.255k
OperatorCostMatched/target_cost:6760000/Aggregation/13255/real_time_st
ddev              14085 ns          12792 ns

3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:6760000/Aggregation/13255/real_time_cv
                0.32 %           0.29 %           3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:6760000/HashJoin/17114/17114/real_time
_mean            3043218 ns        3022074 ns

3 lhs_rows=17.114k model_cost=6.76003M output_rows=17.114k
plan_cost=10.1828M rhs_rows=17.114k
OperatorCostMatched/target_cost:6760000/HashJoin/17114/17114/real_time
_median          3034737 ns        3013058 ns

3 lhs_rows=17.114k model_cost=6.76003M output_rows=17.114k
plan_cost=10.1828M rhs_rows=17.114k
OperatorCostMatched/target_cost:6760000/HashJoin/17114/17114/real_time
_stddev          25639 ns          24888 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:6760000/HashJoin/17114/17114/real_time
_cv              0.84 %           0.82 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:6760000/NestedLoopJoin/311/311/real_ti
me_mean          5812015 ns        5788677 ns

3 lhs_rows=311 model_cost=6.77047M output_rows=311
plan_cost=6.83267M rhs_rows=311
OperatorCostMatched/target_cost:6760000/NestedLoopJoin/311/311/real_ti
me_median        5776632 ns        5752852 ns

3 lhs_rows=311 model_cost=6.77047M output_rows=311
plan_cost=6.83267M rhs_rows=311

```

```

OperatorCostMatched/target_cost:6760000/NestedLoopJoin/311/311/real_ti
me_stddev          65428 ns          65349 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:6760000/NestedLoopJoin/311/311/real_ti
me_cv              1.13 %           1.13 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:6760000/NestedLoopCrossJoin/255/255/re
al_time_mean       5684609 ns       5595092 ns
3 lhs_rows=255 model_cost=6.7626M output_rows=65.025k
plan_cost=6.8136M rhs_rows=255
OperatorCostMatched/target_cost:6760000/NestedLoopCrossJoin/255/255/re
al_time_median     5585757 ns       5512270 ns
3 lhs_rows=255 model_cost=6.7626M output_rows=65.025k
plan_cost=6.8136M rhs_rows=255
OperatorCostMatched/target_cost:6760000/NestedLoopCrossJoin/255/255/re
al_time_stddev     283233 ns        295277 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:6760000/NestedLoopCrossJoin/255/255/re
al_time_cv         4.98 %           5.28 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:12250000/SeqScan/122500/real_time_mean
7200013 ns         7134821 ns         3
model_cost=12.25M output_rows=122.5k plan_cost=12.25M rows=122.5k
OperatorCostMatched/target_cost:12250000/SeqScan/122500/real_time_medi
an                 7204414 ns        7137446 ns
3 model_cost=12.25M output_rows=122.5k plan_cost=12.25M
rows=122.5k
OperatorCostMatched/target_cost:12250000/SeqScan/122500/real_time_stdd
ev                91581 ns          86674 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:12250000/SeqScan/122500/real_time_cv
1.27 %            1.21 %             3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:12250000/Filter/122500/real_time_mean
9183200 ns        9123227 ns          3

```

```

model_cost=12.25M output_rows=61.44k plan_cost=24.5M rows=122.5k
OperatorCostMatched/target_cost:12250000/Filter/122500/real_time_media
n          9274288 ns          9209433 ns

3 model_cost=12.25M output_rows=61.44k plan_cost=24.5M
rows=122.5k
OperatorCostMatched/target_cost:12250000/Filter/122500/real_time_stdde
v          246032 ns          243070 ns

3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:12250000/Filter/122500/real_time_cv
          2.68 %          2.66 %          3

model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:12250000/Projection/556818/real_time_m
ean          76154538 ns          75642732 ns

3 model_cost=12.25M output_rows=556.818k plan_cost=67.9318M
rows=556.818k
OperatorCostMatched/target_cost:12250000/Projection/556818/real_time_m
edian          76053968 ns          75558307 ns

3 model_cost=12.25M output_rows=556.818k plan_cost=67.9318M
rows=556.818k
OperatorCostMatched/target_cost:12250000/Projection/556818/real_time_s
tddev          1296686 ns          1279602 ns

3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:12250000/Projection/556818/real_time_c
v          1.70 %          1.69 %

3 model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:12250000/Sort/65536/real_time_mean
          21631748 ns          21479851 ns          3

model_cost=12.2552M output_rows=65.536k plan_cost=18.8088M
rows=65.536k
OperatorCostMatched/target_cost:12250000/Sort/65536/real_time_median
          21592685 ns          21449075 ns          3

model_cost=12.2552M output_rows=65.536k plan_cost=18.8088M
rows=65.536k
OperatorCostMatched/target_cost:12250000/Sort/65536/real_time_stddev
          209379 ns          211435 ns          3

model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:12250000/Sort/65536/real_time_cv
          0.97 %          0.98 %          3

```



```

model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:12250000/Aggregation/24020/real_time_m
ean          10929372 ns      10855646 ns

3 model_cost=12.2502M output_rows=24.02k plan_cost=14.6522M
rows=24.02k
OperatorCostMatched/target_cost:12250000/Aggregation/24020/real_time_m
edian        11005949 ns      10931545 ns

3 model_cost=12.2502M output_rows=24.02k plan_cost=14.6522M
rows=24.02k
OperatorCostMatched/target_cost:12250000/Aggregation/24020/real_time_s
tddev        184863 ns       178071 ns

3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:12250000/Aggregation/24020/real_time_c
v            1.69 %         1.64 %

3 model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:12250000/HashJoin/31013/31013/real_tim
e_mean       6579551 ns      6525622 ns

3 lhs_rows=31.013k model_cost=12.2501M output_rows=31.013k
plan_cost=18.4527M rhs_rows=31.013k
OperatorCostMatched/target_cost:12250000/HashJoin/31013/31013/real_tim
e_median     6319543 ns      6271821 ns

3 lhs_rows=31.013k model_cost=12.2501M output_rows=31.013k
plan_cost=18.4527M rhs_rows=31.013k
OperatorCostMatched/target_cost:12250000/HashJoin/31013/31013/real_tim
e_stddev     521111 ns       515812 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:12250000/HashJoin/31013/31013/real_tim
e_cv          7.92 %         7.90 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:12250000/NestedLoopJoin/418/418/real_t
ime_mean     10203401 ns      10165100 ns

3 lhs_rows=418 model_cost=12.2307M output_rows=418
plan_cost=12.3143M rhs_rows=418
OperatorCostMatched/target_cost:12250000/NestedLoopJoin/418/418/real_t
ime_median   10207017 ns      10169491 ns

3 lhs_rows=418 model_cost=12.2307M output_rows=418

```

```

plan_cost=12.3143M rhs_rows=418
OperatorCostMatched/target_cost:12250000/NestedLoopJoin/418/418/real_t
ime_stddev          25817 ns          28217 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:12250000/NestedLoopJoin/418/418/real_t
ime_cv              0.25 %          0.28 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:12250000/NestedLoopCrossJoin/343/343/r
eal_time_mean      11246723 ns      11120161 ns

3 lhs_rows=343 model_cost=12.2355M output_rows=117.649k
plan_cost=12.3041M rhs_rows=343
OperatorCostMatched/target_cost:12250000/NestedLoopCrossJoin/343/343/r
eal_time_median    11156642 ns      11021816 ns

3 lhs_rows=343 model_cost=12.2355M output_rows=117.649k
plan_cost=12.3041M rhs_rows=343
OperatorCostMatched/target_cost:12250000/NestedLoopCrossJoin/343/343/r
eal_time_stddev    358953 ns        352800 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:12250000/NestedLoopCrossJoin/343/343/r
eal_time_cv        3.19 %          3.17 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:26010000/SeqScan/260100/real_time_mean
17194941 ns        17060021 ns        3
model_cost=26.01M output_rows=260.1k plan_cost=26.01M rows=260.1k
OperatorCostMatched/target_cost:26010000/SeqScan/260100/real_time_medi
an                  17199810 ns      17065589 ns

3 model_cost=26.01M output_rows=260.1k plan_cost=26.01M
rows=260.1k
OperatorCostMatched/target_cost:26010000/SeqScan/260100/real_time_stddev
ev                  69765 ns        71670 ns

3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:26010000/SeqScan/260100/real_time_cv
0.41 %             0.42 %             3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%

```

```

OperatorCostMatched/target_cost:26010000/Filter/260100/real_time_mean
    20268578 ns    20143106 ns    3
model_cost=26.01M output_rows=130.052k plan_cost=52.02M rows=260.1k
OperatorCostMatched/target_cost:26010000/Filter/260100/real_time_media
n
    20226671 ns    20105167 ns
3 model_cost=26.01M output_rows=130.052k plan_cost=52.02M
rows=260.1k
OperatorCostMatched/target_cost:26010000/Filter/260100/real_time_stdde
v
    948813 ns    923956 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:26010000/Filter/260100/real_time_cv
    4.68 %    4.59 %    3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:26010000/Projection/1182273/real_time_
mean
    158375198 ns    157223598 ns
3 model_cost=26.01M output_rows=1.18227M plan_cost=144.237M
rows=1.18227M
OperatorCostMatched/target_cost:26010000/Projection/1182273/real_time_
median
    158155639 ns    157165675 ns
3 model_cost=26.01M output_rows=1.18227M plan_cost=144.237M
rows=1.18227M
OperatorCostMatched/target_cost:26010000/Projection/1182273/real_time_
stddev
    1530329 ns    1447113 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:26010000/Projection/1182273/real_time_
cv
    0.97 %    0.92 %
3 model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:26010000/Sort/131364/real_time_mean
    66118518 ns    65605278 ns    3
model_cost=26.0101M output_rows=131.364k plan_cost=39.1465M
rows=131.364k
OperatorCostMatched/target_cost:26010000/Sort/131364/real_time_median
    65766383 ns    65262983 ns    3
model_cost=26.0101M output_rows=131.364k plan_cost=39.1465M
rows=131.364k
OperatorCostMatched/target_cost:26010000/Sort/131364/real_time_stddev
    1233820 ns    1224222 ns    3
model_cost=0 output_rows=0 plan_cost=0 rows=0

```

OperatorCostMatched/target_cost:26010000/Sort/131364/real_time_cv
1.87 % 1.87 % 3
model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:26010000/Aggregation/51000/real_time_m
ean 36824897 ns 36546612 ns
3 model_cost=26.01M output_rows=51k plan_cost=31.11M rows=51k
OperatorCostMatched/target_cost:26010000/Aggregation/51000/real_time_m
edian 37531672 ns 37260669 ns
3 model_cost=26.01M output_rows=51k plan_cost=31.11M rows=51k
OperatorCostMatched/target_cost:26010000/Aggregation/51000/real_time_s
tddev 1680229 ns 1661849 ns
3 model_cost=0 output_rows=0 plan_cost=0 rows=0
OperatorCostMatched/target_cost:26010000/Aggregation/51000/real_time_c
v 4.56 % 4.55 %
3 model_cost=0.00% output_rows=0.00% plan_cost=0.00% rows=0.00%
OperatorCostMatched/target_cost:26010000/HashJoin/65848/65848/real_tim
e_mean 15544874 ns 15433459 ns
3 lhs_rows=65.848k model_cost=26.01M output_rows=65.848k
plan_cost=39.1796M rhs_rows=65.848k
OperatorCostMatched/target_cost:26010000/HashJoin/65848/65848/real_tim
e_median 15391909 ns 15276462 ns
3 lhs_rows=65.848k model_cost=26.01M output_rows=65.848k
plan_cost=39.1796M rhs_rows=65.848k
OperatorCostMatched/target_cost:26010000/HashJoin/65848/65848/real_tim
e_stddev 395978 ns 386476 ns
3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:26010000/HashJoin/65848/65848/real_tim
e_cv 2.55 % 2.50 %
3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:26010000/NestedLoopJoin/610/610/real_t
ime_mean 21482184 ns 21403674 ns
3 lhs_rows=610 model_cost=26.047M output_rows=610
plan_cost=26.169M rhs_rows=610

```

OperatorCostMatched/target_cost:26010000/NestedLoopJoin/610/610/real_t
ime_median          21404013 ns      21323764 ns

3 lhs_rows=610 model_cost=26.047M output_rows=610
plan_cost=26.169M rhs_rows=610
OperatorCostMatched/target_cost:26010000/NestedLoopJoin/610/610/real_t
ime_stddev          178260 ns       184469 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:26010000/NestedLoopJoin/610/610/real_t
ime_cv              0.83 %         0.86 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%
OperatorCostMatched/target_cost:26010000/NestedLoopCrossJoin/500/500/r
eal_time_mean       24928068 ns     24677118 ns

3 lhs_rows=500 model_cost=26M output_rows=250k plan_cost=26.1M
rhs_rows=500
OperatorCostMatched/target_cost:26010000/NestedLoopCrossJoin/500/500/r
eal_time_median     24799747 ns     24566264 ns

3 lhs_rows=500 model_cost=26M output_rows=250k plan_cost=26.1M
rhs_rows=500
OperatorCostMatched/target_cost:26010000/NestedLoopCrossJoin/500/500/r
eal_time_stddev     300738 ns       283592 ns

3 lhs_rows=0 model_cost=0 output_rows=0 plan_cost=0 rhs_rows=0
OperatorCostMatched/target_cost:26010000/NestedLoopCrossJoin/500/500/r
eal_time_cv         1.21 %         1.15 %

3 lhs_rows=0.00% model_cost=0.00% output_rows=0.00%
plan_cost=0.00% rhs_rows=0.00%

```

```

[6]: import json
import re
from pathlib import Path

import matplotlib.pyplot as plt
import pandas as pd

RESULTS = Path("benchmark-results")
UNIT_TO_MS = {"ns": 1e-6, "us": 1e-3, "ms": 1.0, "s": 1e3}

def mean_rows(filename):
    data = json.loads((RESULTS / filename).read_text())

```

```

    return [row for row in data["benchmarks"] if row.get("aggregate_name") == "mean"]

```

```

def time_ms(row):
    return row["real_time"] * UNIT_TO_MS[row["time_unit"]]

```

```

[7]: synthetic_pattern = re.compile(
    r"BM_SQL<(P<executor>[^,]+), (P<query>[^,]+), PlannerMode::k(?
    P<mode>[^>+)>/real_time_mean"
)
synthetic = []
for row in mean_rows("query.json"):
    match = synthetic_pattern.fullmatch(row["name"])
    synthetic.append(**match.groupdict(), "time_ms": time_ms(row))
synthetic = pd.DataFrame(synthetic)

ssb = []
for row in mean_rows("ssb-sf001.json"):
    _, query, executor, mode, _ = row["name"].split("/")
    ssb.append({"query": query, "executor": executor, "mode": mode, "time_ms":
    time_ms(row)})
ssb = pd.DataFrame(ssb)

synthetic, ssb

```

```

[7]: (
      executor      query      mode \
0      InterpretedExpressionExecutor  kSimpleSelectSmall      Naive
1      InterpretedExpressionExecutor  kSimpleSelectSmall  Optimized
2  CachedJitCompiledExpressionExecutor  kSimpleSelectSmall      Naive
3  CachedJitCompiledExpressionExecutor  kSimpleSelectSmall  Optimized
4      InterpretedExpressionExecutor      kJoinSmall      Naive
5      InterpretedExpressionExecutor      kJoinSmall  Optimized
6  CachedJitCompiledExpressionExecutor      kJoinSmall      Naive
7  CachedJitCompiledExpressionExecutor      kJoinSmall  Optimized
8      InterpretedExpressionExecutor      kComplex5      Naive
9      InterpretedExpressionExecutor      kComplex5  Optimized
10  CachedJitCompiledExpressionExecutor      kComplex5      Naive
11  CachedJitCompiledExpressionExecutor      kComplex5  Optimized

      time_ms
0      0.022492
1      0.020696
2      0.020039
3      0.019665
4      0.071494
5      0.071193
6      0.074523

```

```

7  0.070604
8  0.088746
9  0.093122
10 0.084456
11 0.080405 ,
    query      executor      mode      time_ms
0  q2.1  Interpreted      Naive    9.535917
1  q2.1  Interpreted  Optimized 10.018523
2  q1.3  Interpreted      Naive    8.894810
3  q1.3  Interpreted  Optimized  7.216482
4  q1.2  Interpreted      Naive    8.755501
5  q1.2  Interpreted  Optimized  6.842163
6  q3.3  Interpreted      Naive   38.770551
7  q3.3  Interpreted  Optimized  6.165384
8  q3.4  Interpreted      Naive   38.014861
9  q3.4  Interpreted  Optimized  6.548677
10 q4.3  Interpreted      Naive    9.839082
11 q4.3  Interpreted  Optimized  6.726511
12 q1.1  Interpreted      Naive    8.337287
13 q1.1  Interpreted  Optimized  7.009090
14 q3.2  Interpreted      Naive   41.168218
15 q3.2  Interpreted  Optimized  7.331045
16 q2.2  Interpreted      Naive   10.611863
17 q2.2  Interpreted  Optimized  7.990721
18 q3.1  Interpreted      Naive   41.047048
19 q3.1  Interpreted  Optimized  7.329905
20 q4.2  Interpreted      Naive   11.383627
21 q4.2  Interpreted  Optimized 13.753622
22 q4.1  Interpreted      Naive    9.885573
23 q4.1  Interpreted  Optimized 13.621071
24 q2.3  Interpreted      Naive    9.506406
25 q2.3  Interpreted  Optimized  6.827825)

```

```

[8]: operator = []
for row in mean_rows("operator-cost.json"):
    parts = row["name"].split("/")
    operator.append({
        "operator": parts[1],
        "input_rows": max(map(int, parts[2:-1])),
        "model_cost": row["model_cost"],
        "time_ms": time_ms(row),
    })
operator = pd.DataFrame(operator)

matched = []
for row in mean_rows("operator-cost-matched.json"):
    parts = row["name"].split("/")

```

```

    matched.append({
        "target_cost": int(parts[1].split(":")[1]),
        "operator": parts[2],
        "input_rows": max(map(int, parts[3:-1])),
        "model_cost": row["model_cost"],
        "time_ms": time_ms(row),
    })
matched = pd.DataFrame(matched)

operator, matched

```

```

[8]: (
      operator  input_rows  model_cost  time_ms
0      SeqScan         1024    102400.0   0.038473
1      SeqScan         2048    204800.0   0.069707
2      SeqScan         4096    409600.0   0.154163
3      SeqScan         8192    819200.0   0.390148
4      SeqScan        16384   1638400.0   0.924368
..      ...
59  NestedLoopCrossJoin         64    425984.0   0.288639
60  NestedLoopCrossJoin        128   1703936.0   0.899354
61  NestedLoopCrossJoin        256   6815744.0   5.036885
62  NestedLoopCrossJoin        512  27262976.0  25.155229
63  NestedLoopCrossJoin       1024 109051904.0 125.736836

```

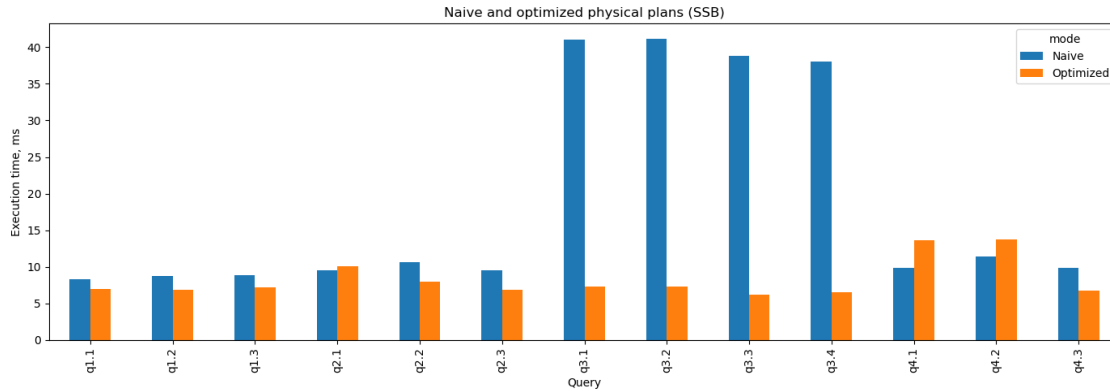
[64 rows x 4 columns],

	target_cost	operator	input_rows	model_cost	time_ms
0	640000	SeqScan	6400	640000.0	0.320159
1	640000	Filter	6400	640000.0	0.409248
2	640000	Projection	29091	640002.0	3.097292
3	640000	Sort	4476	640068.0	0.714273
4	640000	Aggregation	1255	640050.0	0.353749
5	640000	HashJoin	1620	639900.0	0.251418
6	640000	NestedLoopJoin	96	645120.0	0.676351
7	640000	NestedLoopCrossJoin	78	632736.0	0.398744
8	1000000	SeqScan	10000	1000000.0	0.352346
9	1000000	Filter	10000	1000000.0	0.664349
10	1000000	Projection	45455	1000010.0	5.560017
11	1000000	Sort	6993	999999.0	1.181814
12	1000000	Aggregation	1961	1000110.0	0.540896
13	1000000	HashJoin	2532	1000140.0	0.408221
14	1000000	NestedLoopJoin	120	1008000.0	1.006221
15	1000000	NestedLoopCrossJoin	98	998816.0	0.600207
16	3240000	SeqScan	32400	3240000.0	1.284652
17	3240000	Filter	32400	3240000.0	2.176167
18	3240000	Projection	147273	3240006.0	20.058836
19	3240000	Sort	19636	3239940.0	4.184701
20	3240000	Aggregation	6353	3240030.0	1.881536

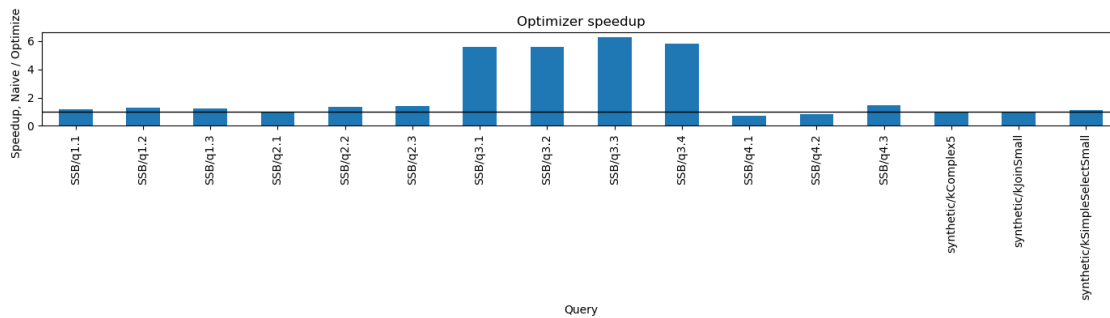
21	3240000	HashJoin	8203	3240185.0	1.306558
22	3240000	NestedLoopJoin	215	3235750.0	2.883331
23	3240000	NestedLoopCrossJoin	177	3258216.0	1.967699
24	6760000	SeqScan	67600	6760000.0	3.804148
25	6760000	Filter	67600	6760000.0	4.921571
26	6760000	Projection	307273	6760006.0	41.998731
27	6760000	Sort	38409	6759984.0	10.271929
28	6760000	Aggregation	13255	6760050.0	4.438157
29	6760000	HashJoin	17114	6760030.0	3.043218
30	6760000	NestedLoopJoin	311	6770470.0	5.812015
31	6760000	NestedLoopCrossJoin	255	6762600.0	5.684609
32	12250000	SeqScan	122500	12250000.0	7.200013
33	12250000	Filter	122500	12250000.0	9.183200
34	12250000	Projection	556818	12249996.0	76.154538
35	12250000	Sort	65536	12255232.0	21.631748
36	12250000	Aggregation	24020	12250200.0	10.929372
37	12250000	HashJoin	31013	12250135.0	6.579551
38	12250000	NestedLoopJoin	418	12230680.0	10.203401
39	12250000	NestedLoopCrossJoin	343	12235496.0	11.246723
40	26010000	SeqScan	260100	26010000.0	17.194941
41	26010000	Filter	260100	26010000.0	20.268578
42	26010000	Projection	1182273	26010006.0	158.375198
43	26010000	Sort	131364	26010072.0	66.118518
44	26010000	Aggregation	51000	26010000.0	36.824897
45	26010000	HashJoin	65848	26009960.0	15.544874
46	26010000	NestedLoopJoin	610	26047000.0	21.482184
47	26010000	NestedLoopCrossJoin	500	26000000.0	24.928068

```
[15]: ssb_interpreted = ssb[ssb.executor == "Interpreted"].copy()

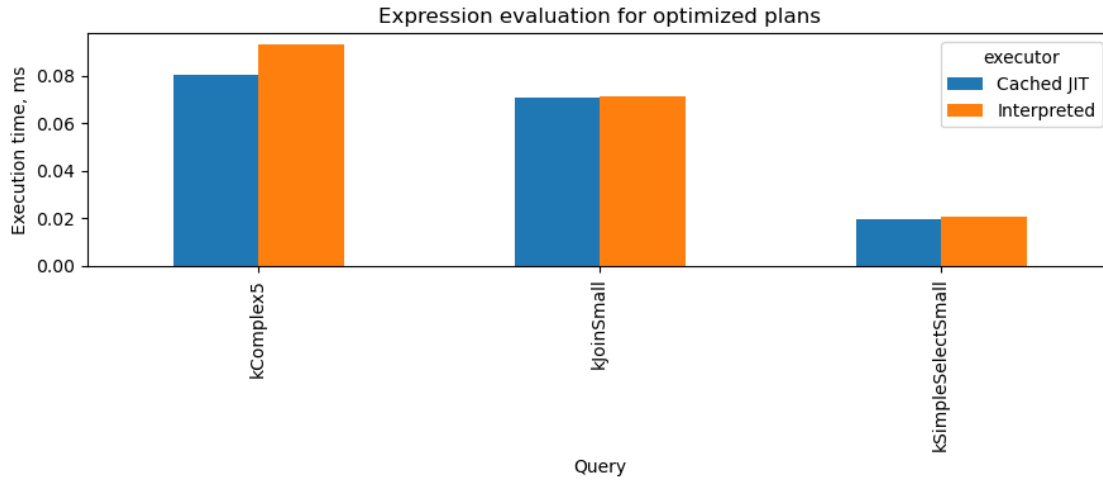
wide = ssb_interpreted.pivot(index="query", columns="mode", values="time_ms").
    ↪sort_index()
ax = wide[["Naive", "Optimized"]].plot.bar(figsize=(14, 5))
ax.set_ylabel("Execution time, ms")
ax.set_xlabel("Query")
ax.set_title("Naive and optimized physical plans (SSB)")
plt.tight_layout()
```



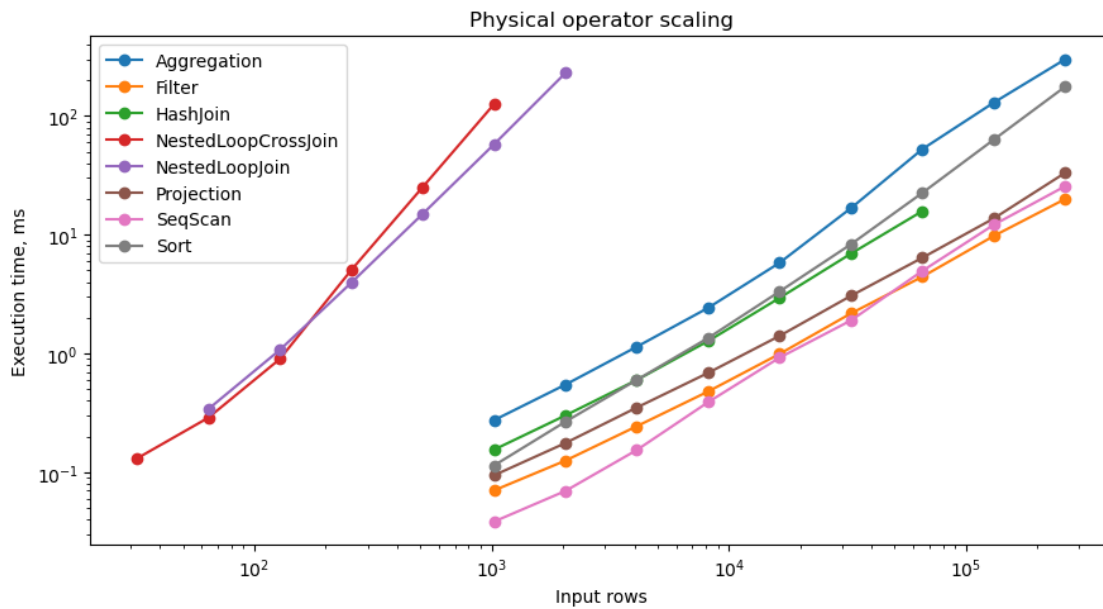
```
[10]: speedup = (wide["Naive"] / wide["Optimized"]).sort_index()
ax = speedup.plot.bar(figsize=(14, 4))
ax.axhline(1.0, color="black", linewidth=1)
ax.set_ylabel("Speedup, Naive / Optimized")
ax.set_xlabel("Query")
ax.set_title("Optimizer speedup")
plt.tight_layout()
```



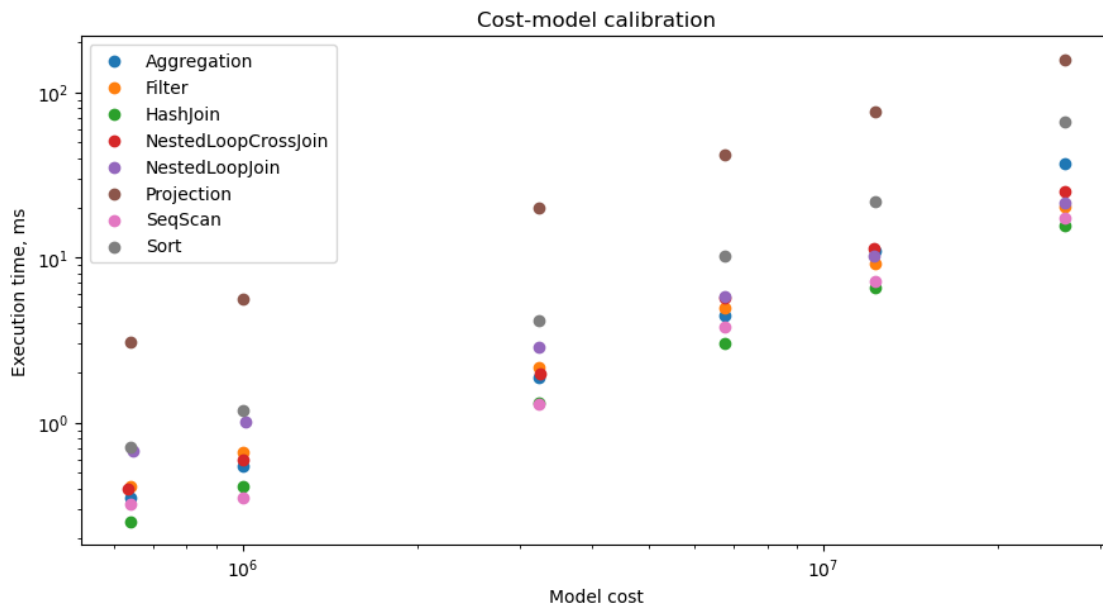
```
[11]: jit = synthetic[synthetic["mode"] == "Optimized"].copy()
jit["executor"] = jit["executor"].replace({
    "InterpretedExpressionExecutor": "Interpreted",
    "CachedJitCompiledExpressionExecutor": "Cached JIT",
})
ax = jit.pivot(index="query", columns="executor", values="time_ms").plot.
    bar(figsize=(9, 4))
ax.set_ylabel("Execution time, ms")
ax.set_xlabel("Query")
ax.set_title("Expression evaluation for optimized plans")
plt.tight_layout()
```



```
[12]: fig, ax = plt.subplots(figsize=(9, 5))
for name, group in operator.groupby("operator"):
    group = group.sort_values("input_rows")
    ax.plot(group.input_rows, group.time_ms, marker="o", label=name)
ax.set_xscale("log")
ax.set_yscale("log")
ax.set_xlabel("Input rows")
ax.set_ylabel("Execution time, ms")
ax.set_title("Physical operator scaling")
ax.legend()
plt.tight_layout()
```



```
[13]: fig, ax = plt.subplots(figsize=(9, 5))
for name, group in matched.groupby("operator"):
    ax.scatter(group.model_cost, group.time_ms, label=name)
ax.set_xscale("log")
ax.set_yscale("log")
ax.set_xlabel("Model cost")
ax.set_ylabel("Execution time, ms")
ax.set_title("Cost-model calibration")
ax.legend()
plt.tight_layout()
```



```
[14]: coefficients = pd.DataFrame([
    ("SeqScan", "row", 100),
    ("Filter", "output row", 100),
    ("Projection", "output row", 22),
    ("Sort", "n * (floor(log2(n)) + 1)", 11),
    ("Aggregation", "input row", 510),
    ("NestedLoopJoin", "input row pair", 70),
    ("NestedLoopCrossJoin", "input row pair", 104),
    ("HashJoin", "build row * width", 100),
    ("HashJoin", "probe row", 35),
    ("HashJoin", "output row * width", 10),
], columns=["operator", "coefficient", "value"])
coefficients
```

```

[14]:
      operator      coefficient  value
0      SeqScan              row    100
1        Filter          output row    100
2      Projection          output row     22
3          Sort  n * (floor(log2(n)) + 1)    11
4      Aggregation          input row    510
5      NestedLoopJoin      input row pair     70
6  NestedLoopCrossJoin      input row pair    104
7          HashJoin      build row * width    100
8          HashJoin          probe row     35
9          HashJoin      output row * width     10

```